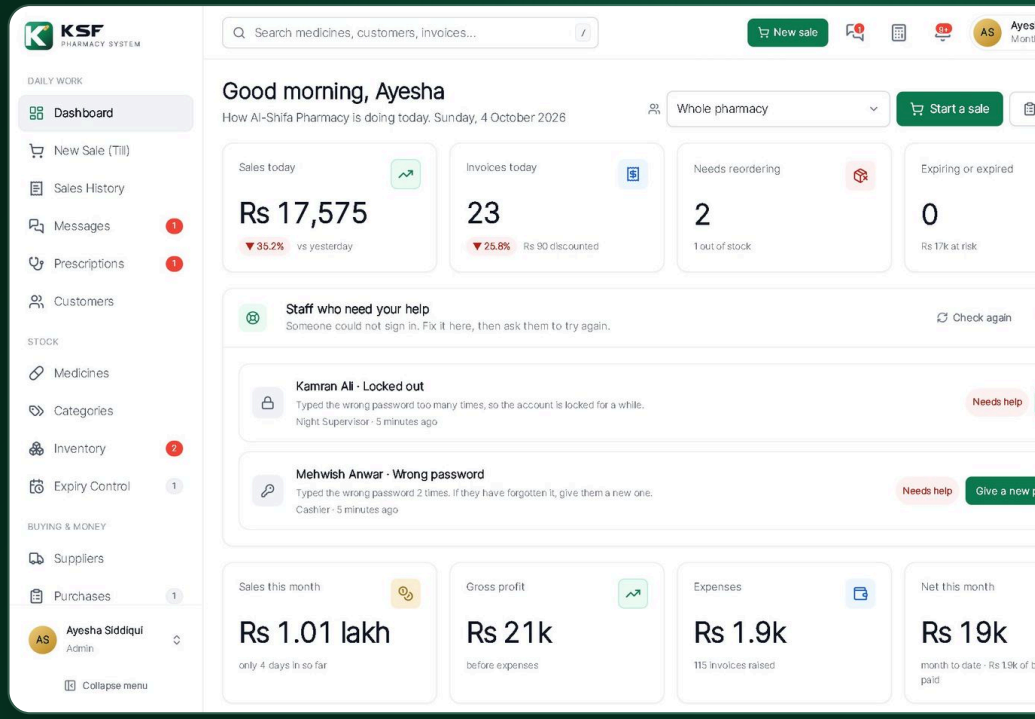
KSF

UPDATE REPORT · VERSION 2.0

KSF Pharmacy Management System

What changed from version 1.0.6, how it looks now, and what you need to do to start.



VERSION
2.0.0

DATE
4 October 2026

PREVIOUS VERSION
1.0.6 (2 October 2026)

BUILT WITH
React and TypeScript · Express and TypeScript · MongoDB

Created by **Saad Bin Saeed Khan** — Software Engineer — MPhil/MS Scholar

4 October 2026

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About the screenshots. Version 2.0 is shown with four sample pharmacies created through the normal sign-up for this report (their sale dates were spread over 30 days so the charts show a month). Version 1.0.6 is shown with its demonstration pharmacy from the previous round.

1 In one page

Everything you asked for in this round, and where to see it in this report.

You asked for	What was done	Section
The whole database on MongoDB	Done. Every pharmacy, sale, plan, payment and referral now lives in one MongoDB database. You paste its link into one line of the .env file.	3
No demonstration data and no demo sign-ins	Done. A new installation holds only the subscription plans and one account: the Super Admin.	4
Only the Super Admin can sign in at the start, and he can change his own e-mail and password	Done. The account is made from .env on the first start. After that it is changed in My account, like any website.	4, 7
One Super Admin over every pharmacy, seeing them together and one by one, with his own income	Done. A separate platform console: overview, pharmacies, income, plans, support desk, people who need help, activity.	7
The Super Admin can sort out any pharmacy's problems	Done. He answers support requests, unlocks people, and can work inside any pharmacy. Everything he does there is recorded.	7
Each pharmacy has its own Admin and its own roles	Done. The person who registers is the pharmacy's Admin and can create roles by ticking what each may do.	5, 8
Keep sign-up, billing, plans, payments and referrals	Kept, all working on MongoDB.	4, 8
Strong sign-in with JWT access and refresh tokens	Done. A 15-minute access token and a refresh token that is replaced every time it is used.	6

You asked for	What was done	Section
A .env file where you add the MongoDB link	Done. Created with fresh secret keys; one line is left for your link.	2, 3
Full width, and right on every screen size	Done. The app uses the whole screen, from small phones to wide monitors.	9
Up to five themes; the Admin decides who may change theirs	Done. Five themes, each with light and dark mode.	10
Every API working, and every View / Edit / Delete window working	Checked: 49 automated tests and 155 end-to-end checks pass, and each screen was opened by hand.	13
Remove files that are not needed; look like professional work	Done. Old demo tools, old drivers, the release archive and out-of-date notes are gone. The folders are explained in section 14.	14
This report in PDF and in Word	This document. The Word file has the same content, written as a proper Word document.	—

Nothing was redesigned

The screens your staff already know look and work the same way. The colours are the same KSF green unless a pharmacy picks another theme. What changed is underneath — the database, sign-in and the new Super Admin console — plus the requests above.

2 Three steps to start

You need Node.js 20 or newer on the computer and a MongoDB database. The free MongoDB Atlas plan is enough to begin.

1. **Put your MongoDB link in .env.** Open the file named .env in the project folder. Find the line MONGODB_URI and paste your connection string between the quotes. DATABASE_SETUP.md shows how to get the link from MongoDB Atlas, click by click.
2. **Start KSF.** On Windows, double-click windows\START_KSF.bat. Or, in a terminal in the project folder, run npm install once and then npm run dev. The browser opens by itself.
3. **Sign in as the Super Admin and change the password.** Use the e-mail and password written in .env (SUPER_ADMIN_EMAIL and SUPER_ADMIN_PASSWORD). Then open My account and change both. From then on, .env is not used for this account any more.

```
# .env — the only line you must fill in
MONGODB_URI="mongodb+srv://ksf_app:YOUR-PASSWORD@cluster0.xxxxx.mongodb.net/?retryWrites=true&w=majority"
MONGODB_DB=ksf_pharmacy
```

Keep .env private

The .env file holds the database password and the keys that sign every sign-in. Do not e-mail it, upload it or share it in a chat. It is already excluded from Git and from Docker images.

To check the link before starting, run **npm run db:check**. It connects and says, line by line, whether everything is ready.

3 The database is now MongoDB

One database for the whole platform. Each pharmacy's records carry its own pharmacy number, and every screen only ever reads its own pharmacy's records.

	Version 1.0.6	Version 2.0
Database	SQLite (a file on one computer) or PostgreSQL	MongoDB — Atlas in the cloud, or MongoDB on your own computer or server
Where the link goes	Several DB_ and PG settings	One line: MONGODB_URI in .env
A new installation contains	A demonstration pharmacy, demo staff and demo sign-in buttons	The four subscription plans and the Super Admin — nothing else
A sale	Saved row by row	Saved as one transaction: the invoice, the stock taken off each batch and the payment are saved together or not at all
Two tills selling the last pack	Guarded in the application	Guarded by the database itself: only one of the two can succeed
Backups	Copy of the database file	npm run db:backup writes a compressed file, then reads it back to prove it is complete. On Atlas, cloud snapshots as well
Checking the setup	npm run db:check	npm run db:check — reports the version, transactions, plans and the Super Admin

Commands for the database

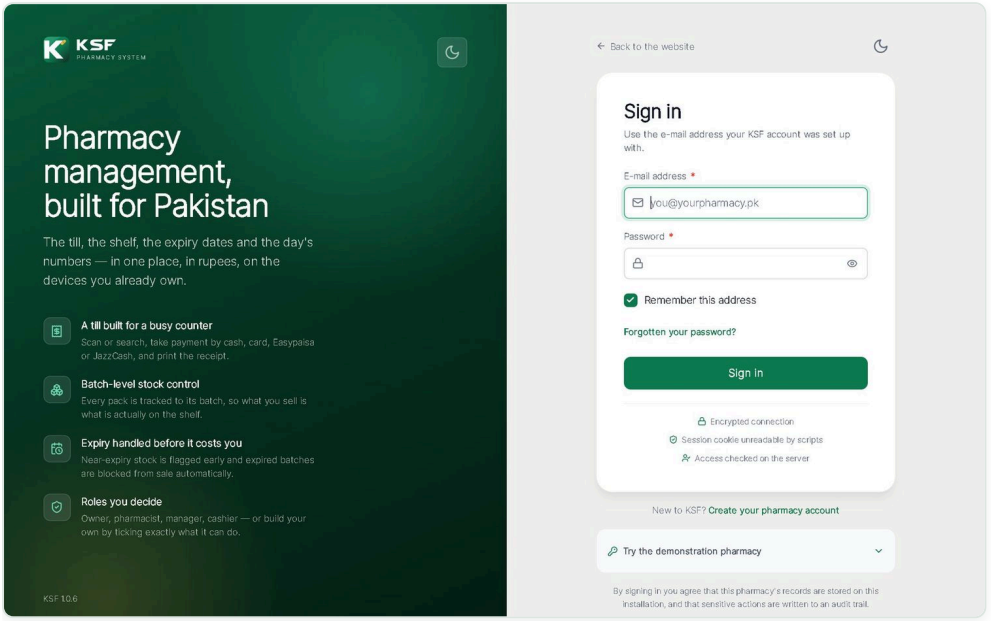
Command	What it does
npm run db:check	Connects and reports in plain words whether everything is ready. Changes nothing.
npm run db:backup	Takes a full backup into .data/backups and checks it can be restored.
npm run db:restore -- <file> --yes	Puts a backup back. The current data is backed up first, so this can be undone too.
npm run db:reset -- --yes	Empties a test database and starts it again. Refuses to run on a live (production) system.
npm run secrets -- --write	Makes new secret keys for sign-in. Doing this signs everybody out.

4 Only the Super Admin at the start

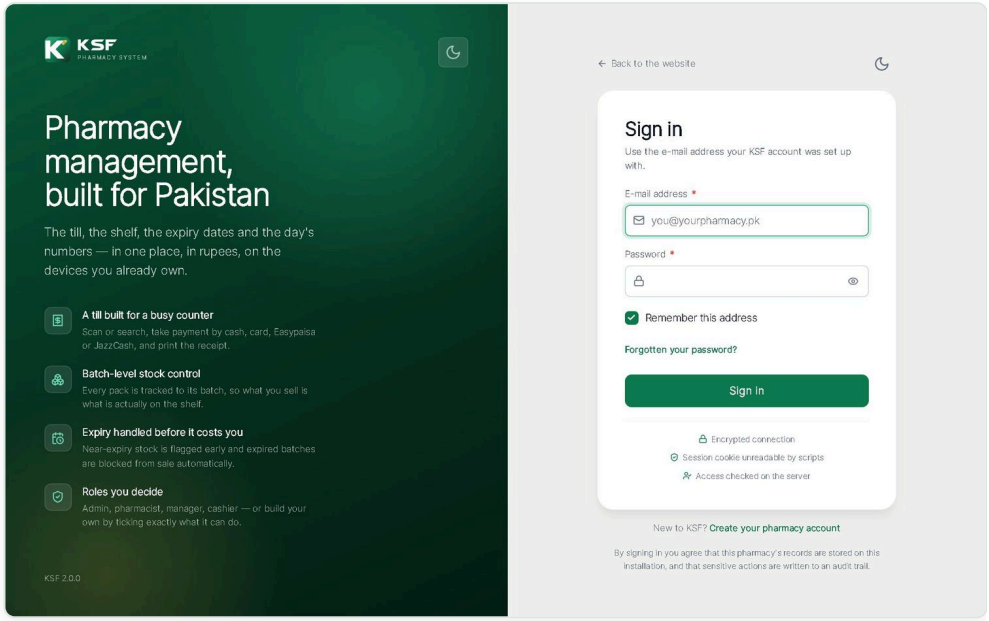
The demonstration pharmacy, the demo staff and the "Try the demonstration pharmacy" buttons are gone. Pharmacies join by signing up on the website.

The sign-in page

PREVIOUS · 1.0.6

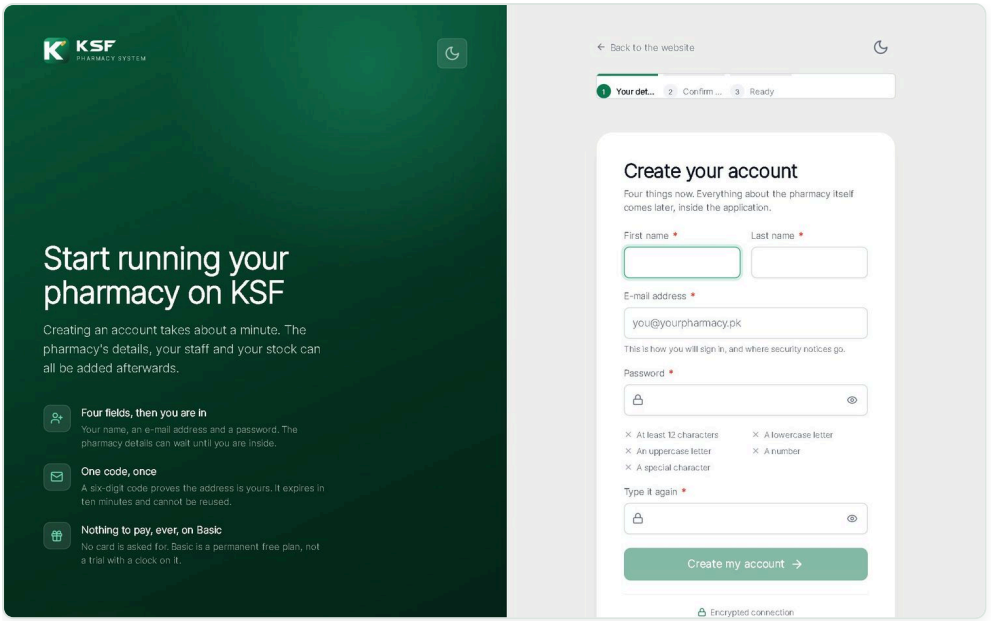


NEW · 2.0

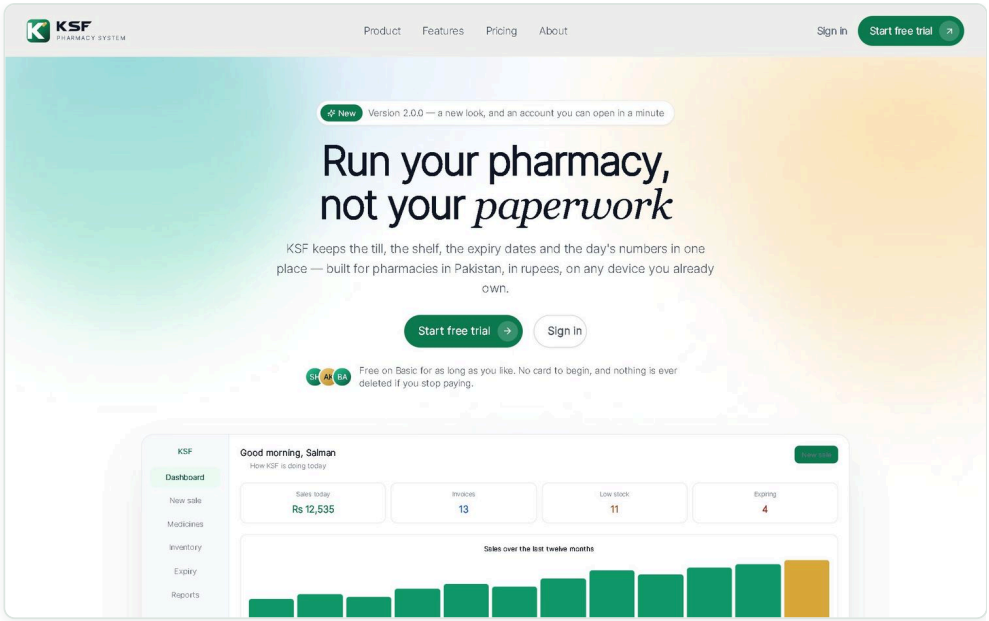


Before: a "Try the demonstration pharmacy" panel offered ready-made demo accounts. Now: an ordinary sign-in page with no shortcuts.

- On its first start, KSF creates the Super Admin from SUPER_ADMIN_EMAIL and SUPER_ADMIN_PASSWORD in .env.
- The Super Admin then changes the e-mail address (confirmed with a code sent to the new address) and the password in **My account**.
- If the Super Admin forgets the password, **Forgotten your password?** on the sign-in page sends a reset code by e-mail.



Sign-up: a new pharmacy creates its account and confirms its e-mail with a code.



The public website is kept, with plans and pricing.

5 Who does what now

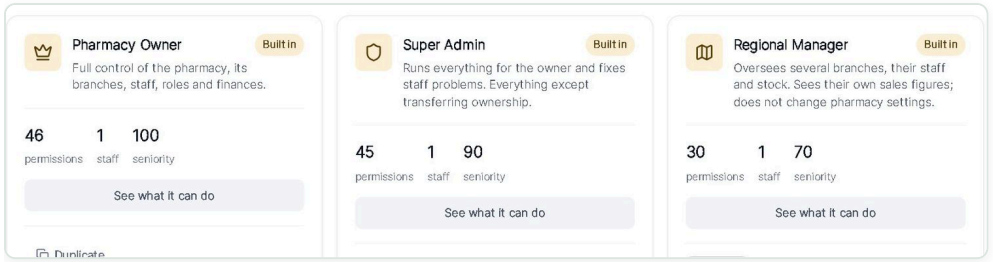
One Super Admin runs the platform. Each pharmacy is run by its own Admin, who decides what the rest of the staff may do.

Who	What they see	What they can do
Super Admin (one for the whole platform)	Every pharmacy, combined and one by one; all income; every support request	Change plans, record payments, suspend and reactivate a pharmacy, answer support, unlock people, work inside any pharmacy
Admin (one per pharmacy — the person who registered it)	Everything in their own pharmacy	Everything in their pharmacy: staff, roles, settings, billing, theme. Creates new roles
Regional Manager	Several branches, their staff and stock	Run stock, buying and the day's numbers; no pharmacy settings
Branch Manager	One branch	Stock, purchasing, refunds, the day close
Pharmacist	Prescriptions, medicines, stock	Verify and dispense prescriptions; look after the catalogue
Cashier	The till and customers	Sell, take payments, record customers

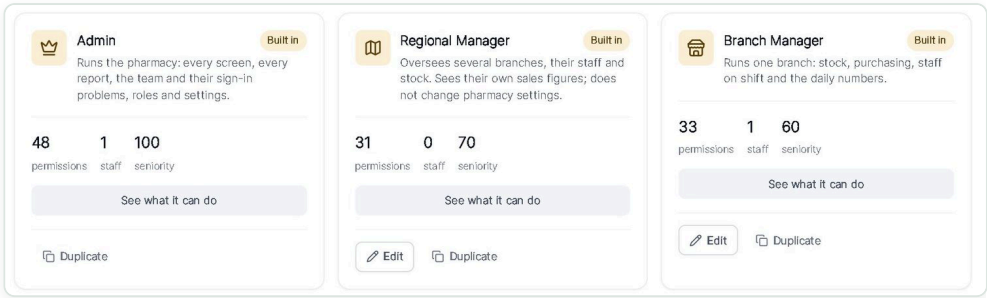
The old **Pharmacy Owner** role is gone: the person who registers a pharmacy is its **Admin**, and above all pharmacies there is now the platform **Super Admin**. A role can never be given more than the person creating it has, and nobody can change someone more senior than themselves. The server checks this on every request — hiding a button is never the only protection.

Roles

PREVIOUS · 1.0.6



NEW · 2.0



Before: "Pharmacy Owner" and "Super Admin" were both roles inside a pharmacy. Now: the pharmacy has an Admin and four more built-in roles, plus any it creates (here, "Night Supervisor").

6 Signing in, in plain words

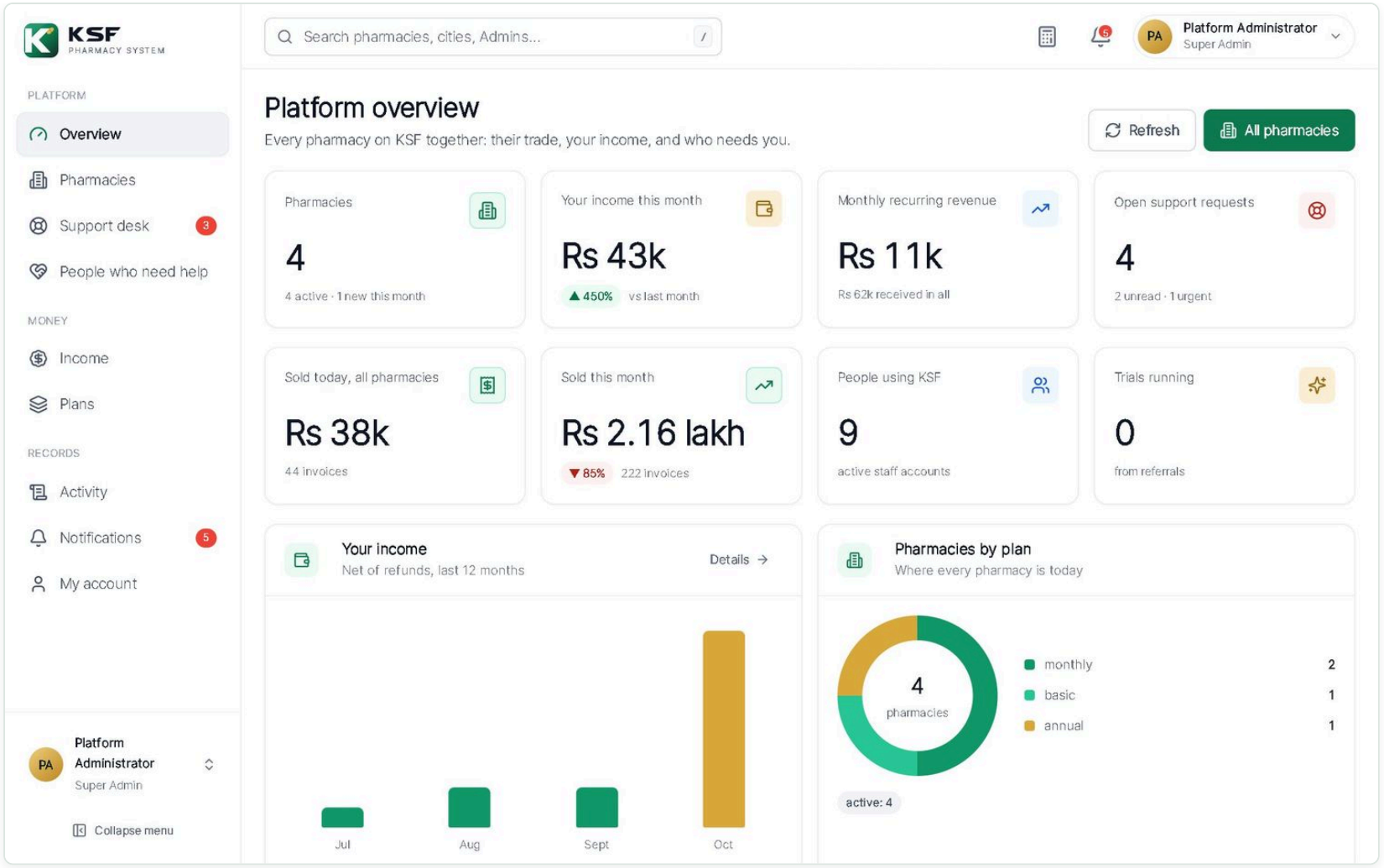
Sign-in now uses two tokens, the way banking and large websites do.

	What it is	Why it matters
Access token	A signed pass that lasts 15 minutes. The browser keeps it in memory only.	If someone copied it, it would be useless within minutes.
Refresh token	A second pass, kept in a protected cookie that page code cannot read. It is replaced with a new one every time it is used.	If an old one is ever presented again, KSF knows it was copied and ends that sign-in at once.
Session check	Every request is checked against the live sign-in record.	Signing out, a password change, suspending a person or removing a device takes effect immediately.
Password again for sensitive actions	Suspending a pharmacy, confirming or refunding a payment, undoing an import, two-step changes, removing a device.	A borrowed, unlocked computer cannot be used for serious changes.

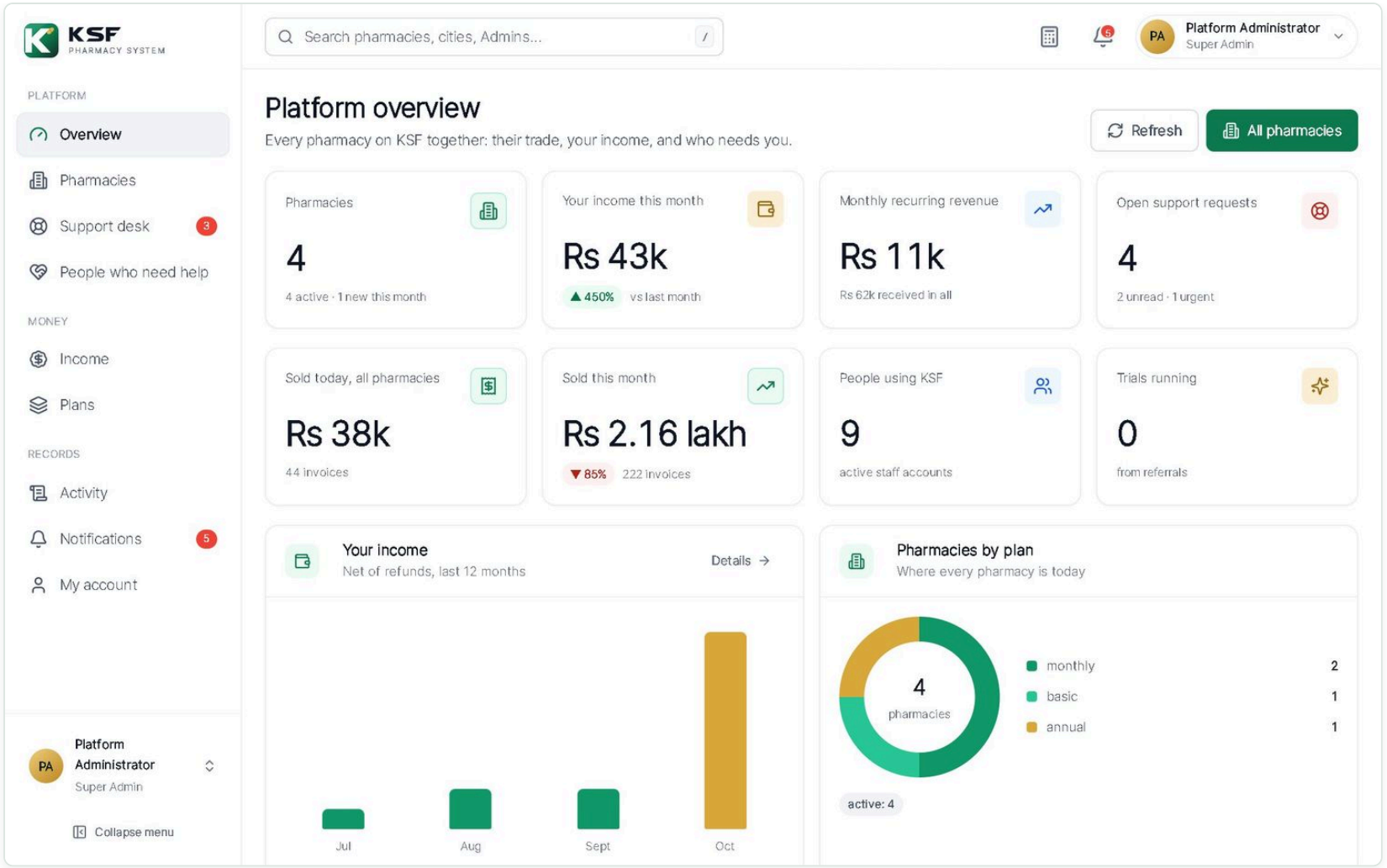
- Passwords are stored as strong one-way hashes (scrypt), at least 12 characters, never written to logs.
- Two-step sign-in with an authenticator app, with one-time recovery codes.
- Accounts lock after repeated wrong passwords; sign-in and sign-up are rate-limited.
- A limit on how many computers and phones each person may use, set by the plan or by the Super Admin.

7 The Super Admin's console

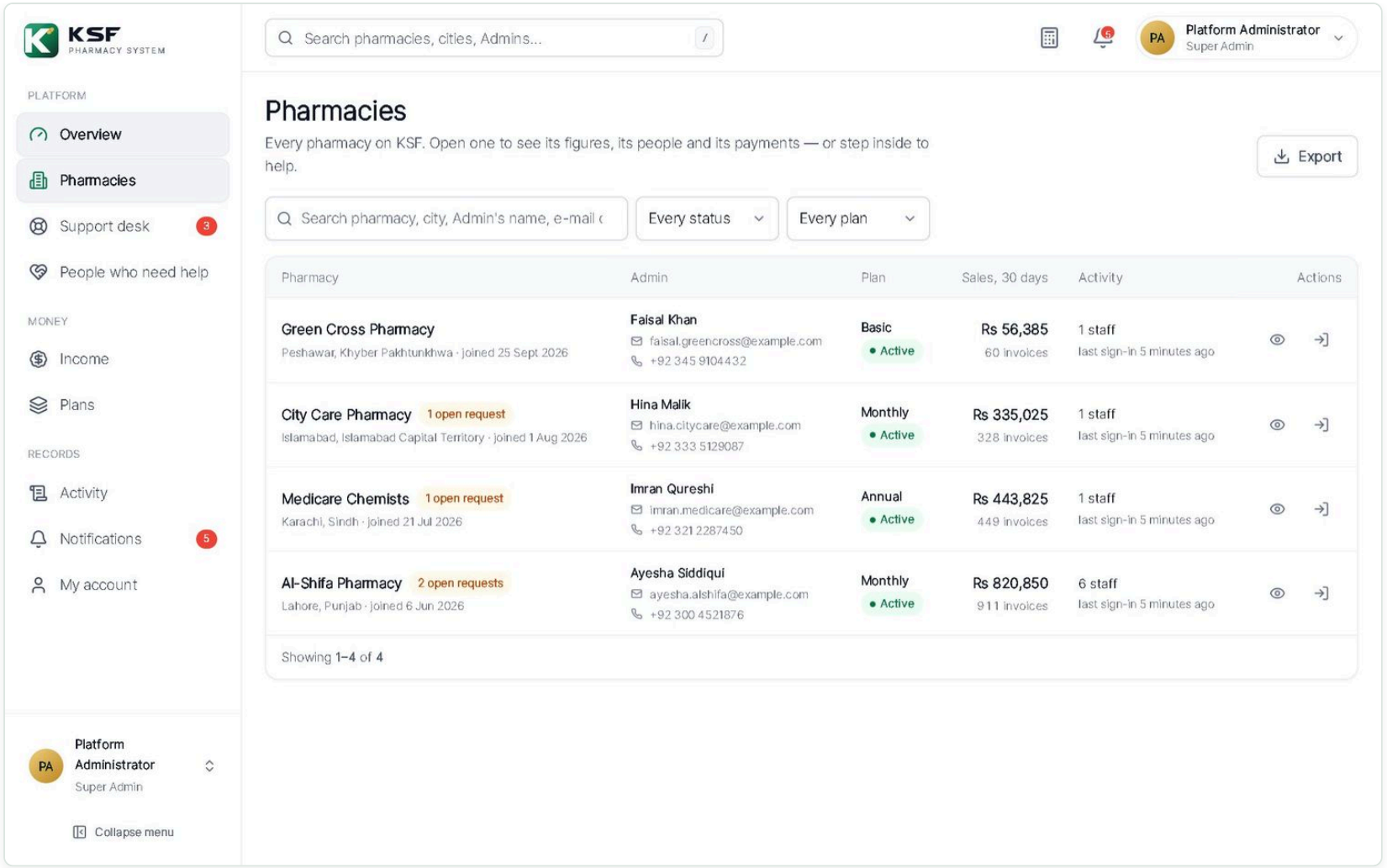
The Super Admin does not belong to any pharmacy. He signs in to a console of his own for the whole platform. The screens below use four sample pharmacies created for this report.



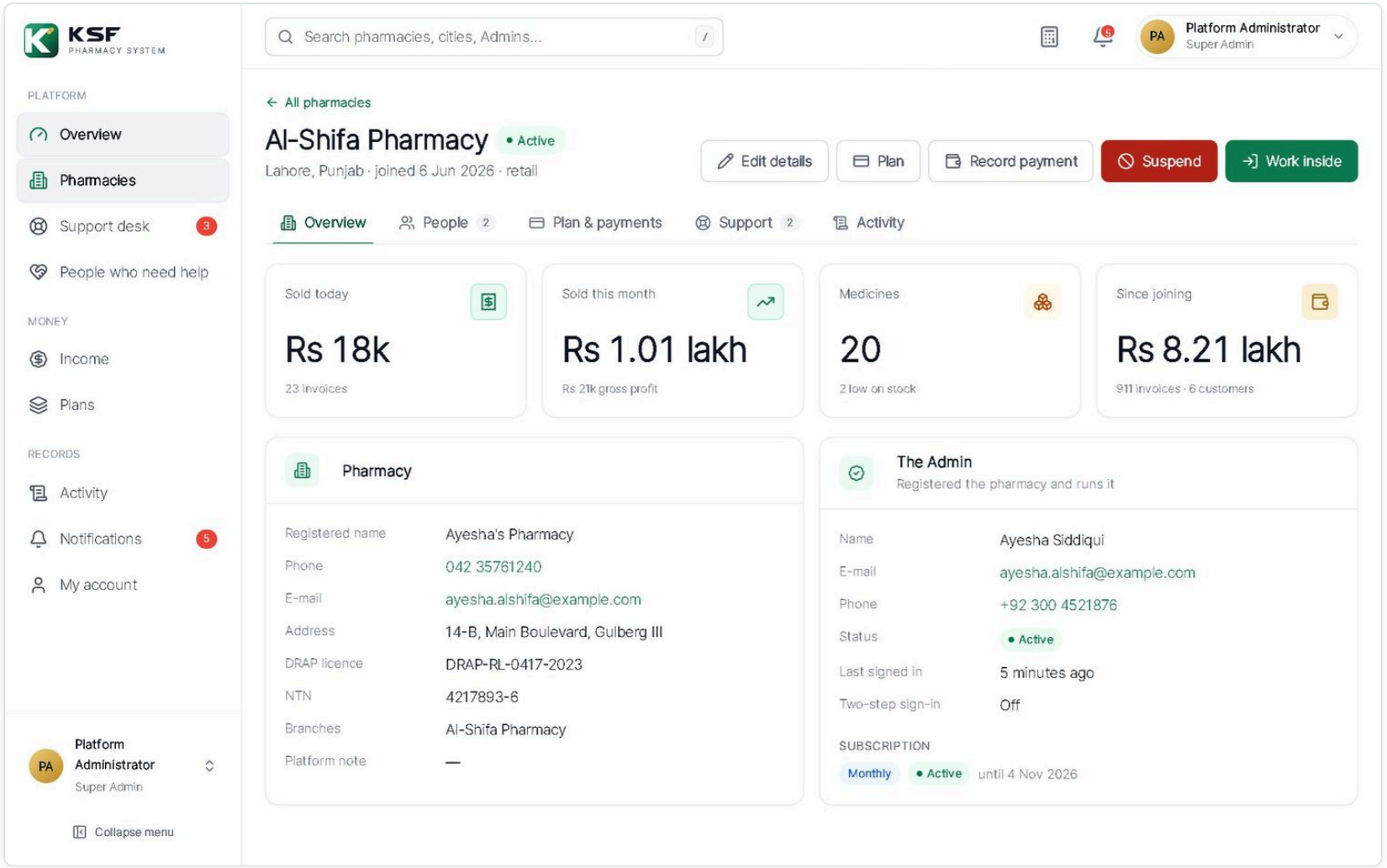
Platform overview: all pharmacies together — how many there are, the Super Admin's income this month, monthly recurring revenue, open support requests, and what all pharmacies sold today and this month.



Further down: income for the last twelve months and pharmacies by plan.



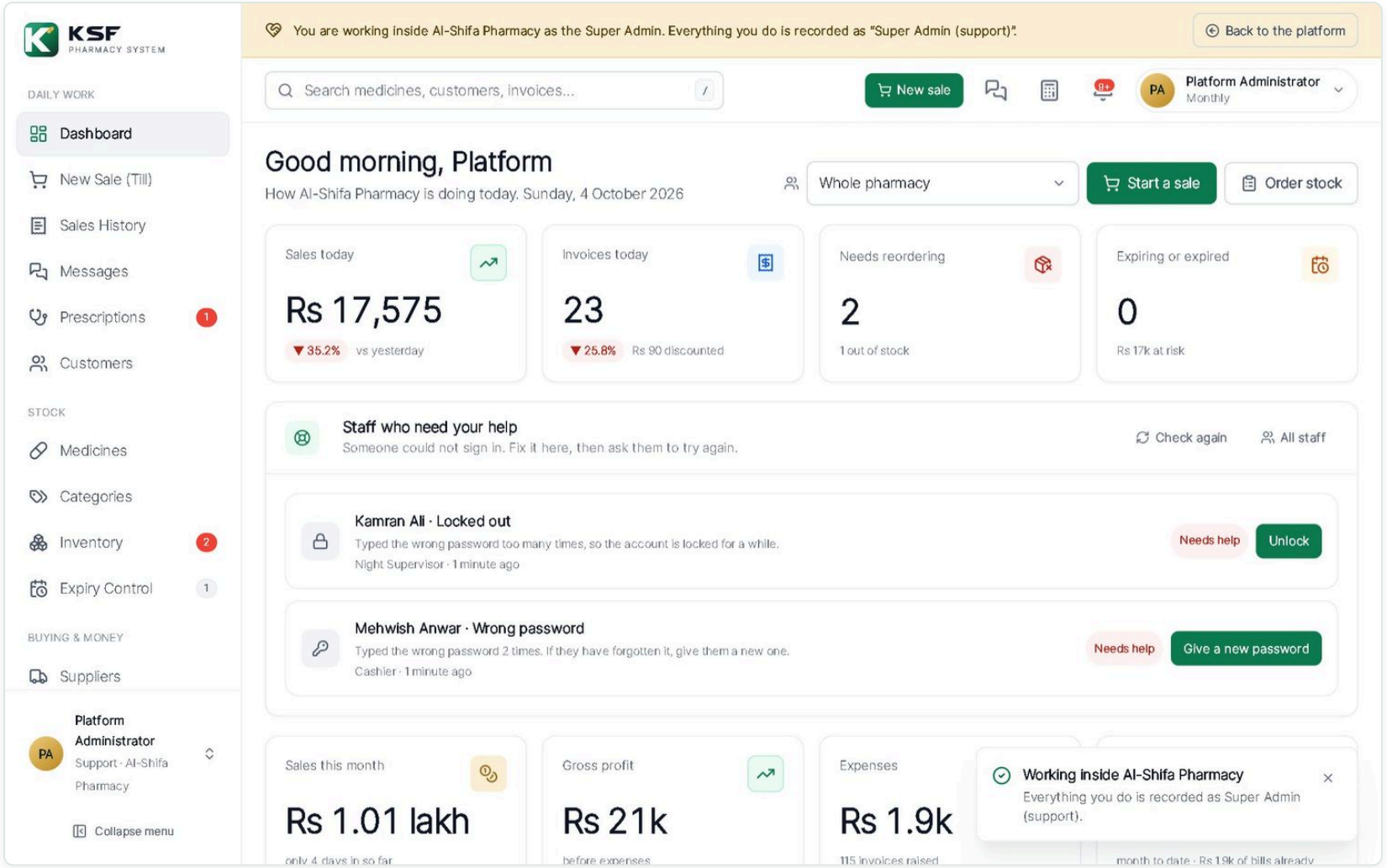
Pharmacies: every pharmacy with its Admin, plan, 30-day sales and activity. Search by name, city or Admin. The eye opens a pharmacy; the arrow works inside it.



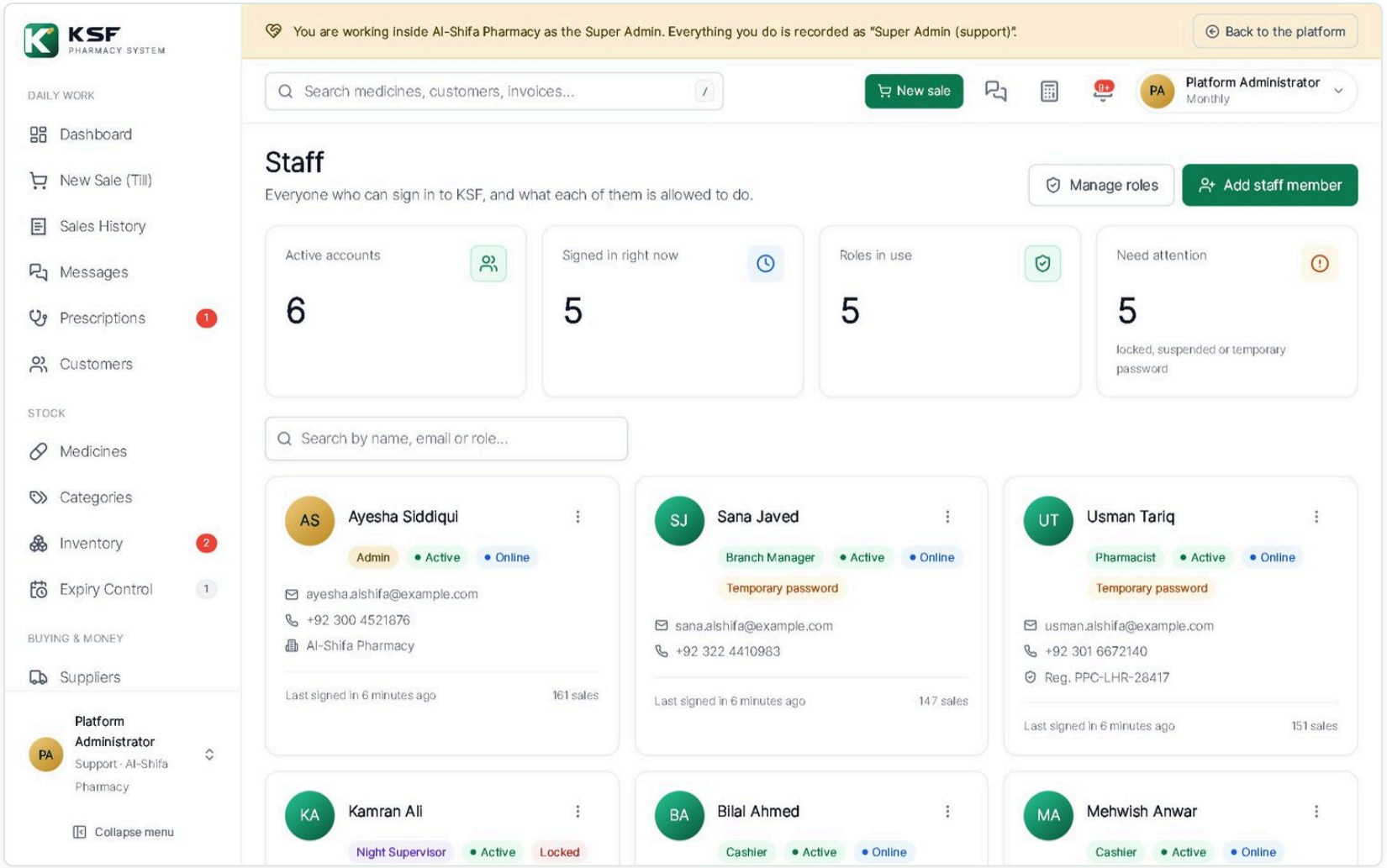
One pharmacy: its figures, details and Admin. Buttons to edit it, change or extend the plan, record a payment, suspend it, or work inside it. Tabs show its people, payments, support requests and activity.

Working inside a pharmacy

Work inside opens the pharmacy as its staff see it, with every permission, so the Super Admin can fix what the pharmacy cannot: unlock a cashier, give someone a new password, correct a role. A banner across the top says whose pharmacy it is, and everything done there is recorded as "Super Admin (support)".

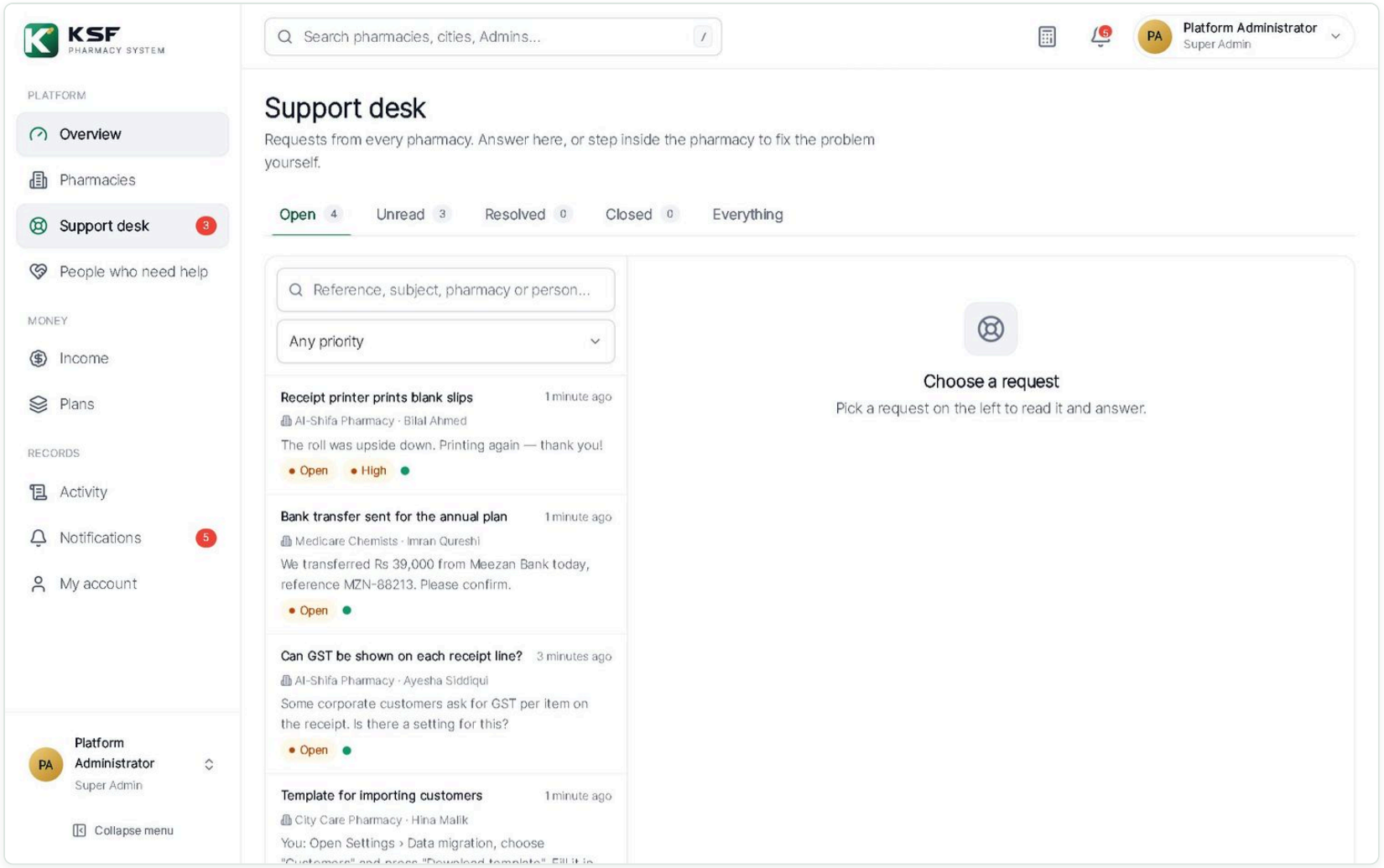


Working inside Al-Shifa Pharmacy. "Back to the platform" returns to the console.

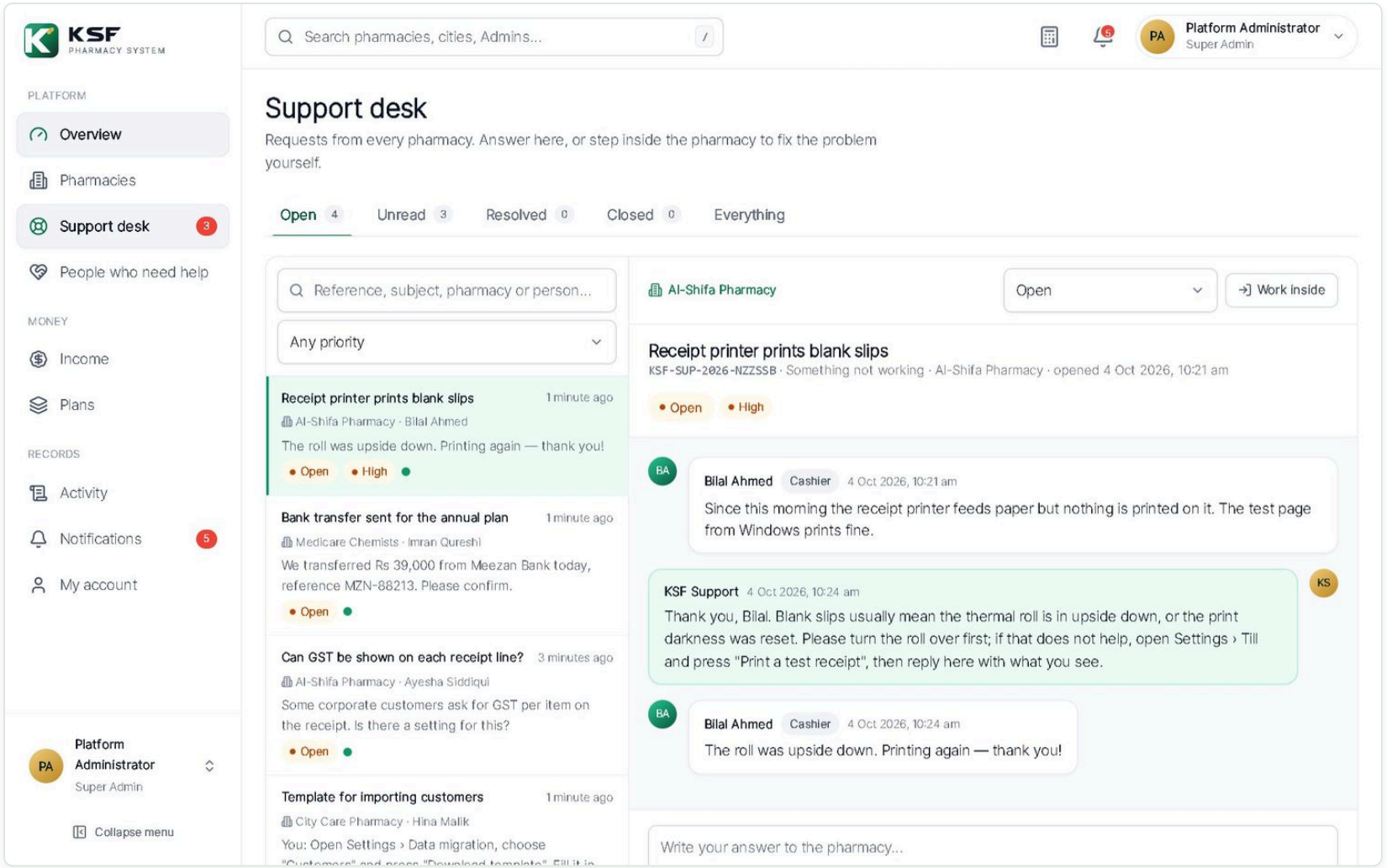


The pharmacy's staff, seen by the Super Admin in support mode.

Support desk

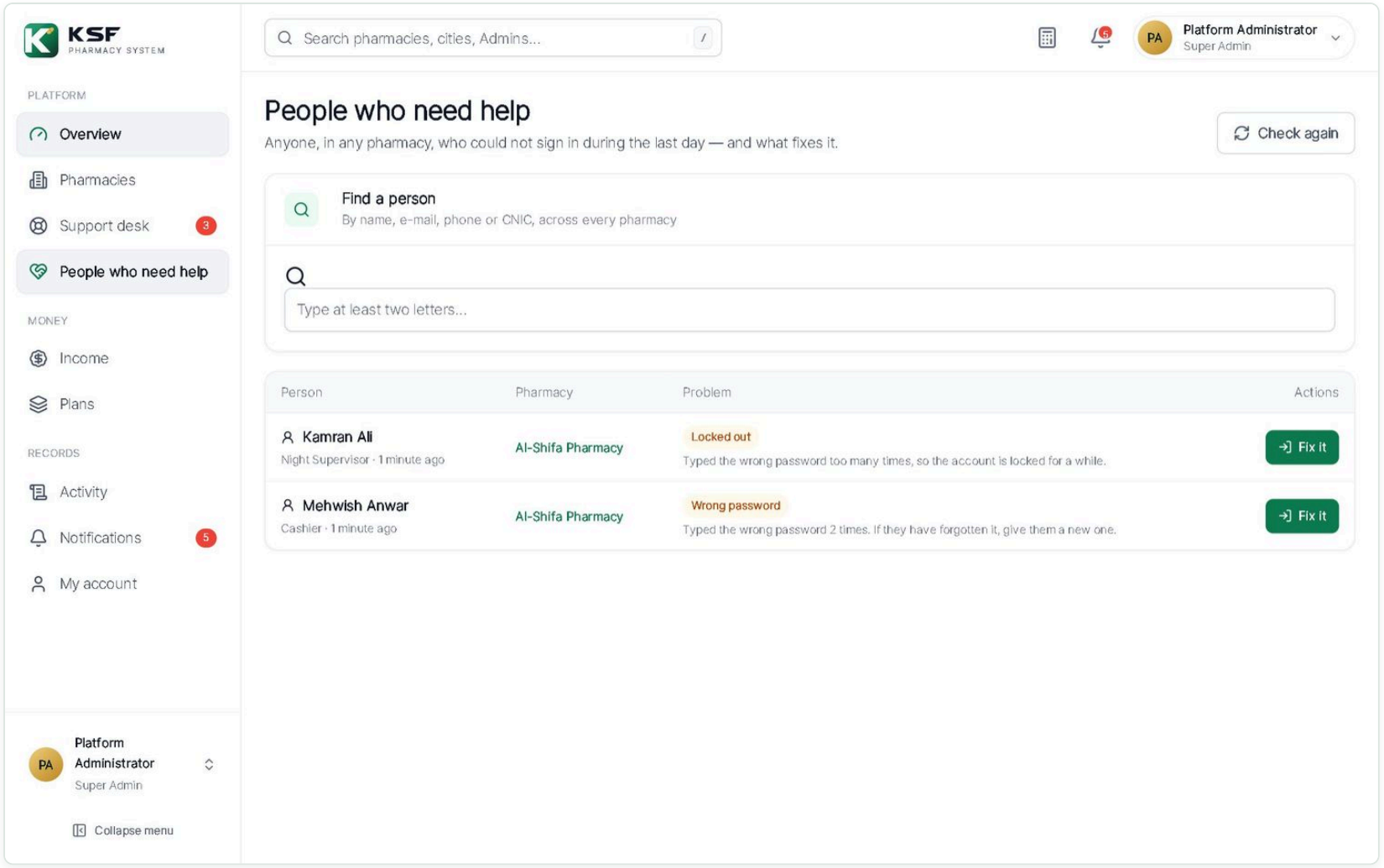


Requests from every pharmacy in one list, newest first, with who raised them and how urgent they are.



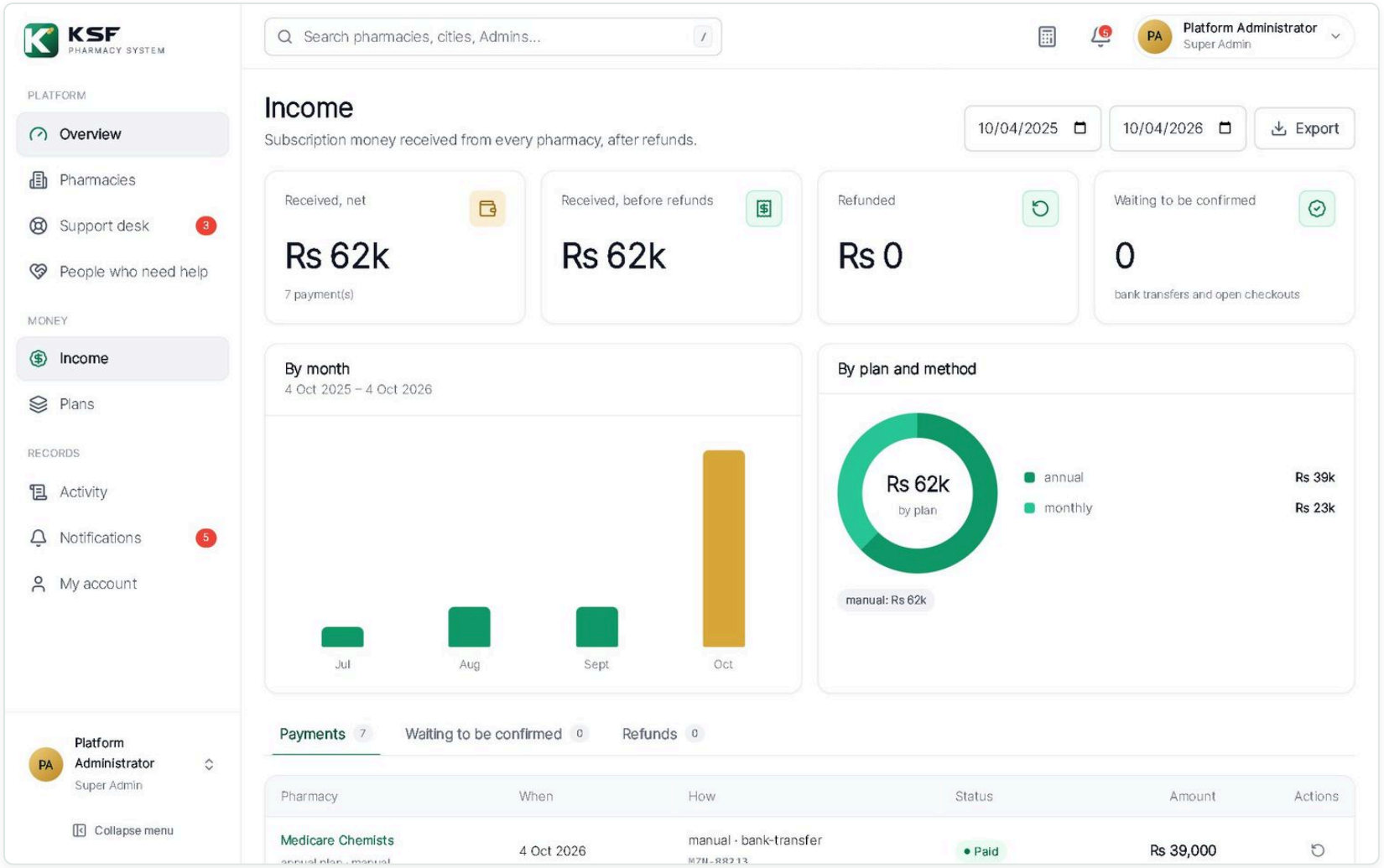
A conversation: the cashier's problem and the Super Admin's answer. The pharmacy is notified when an answer arrives.

People who need help

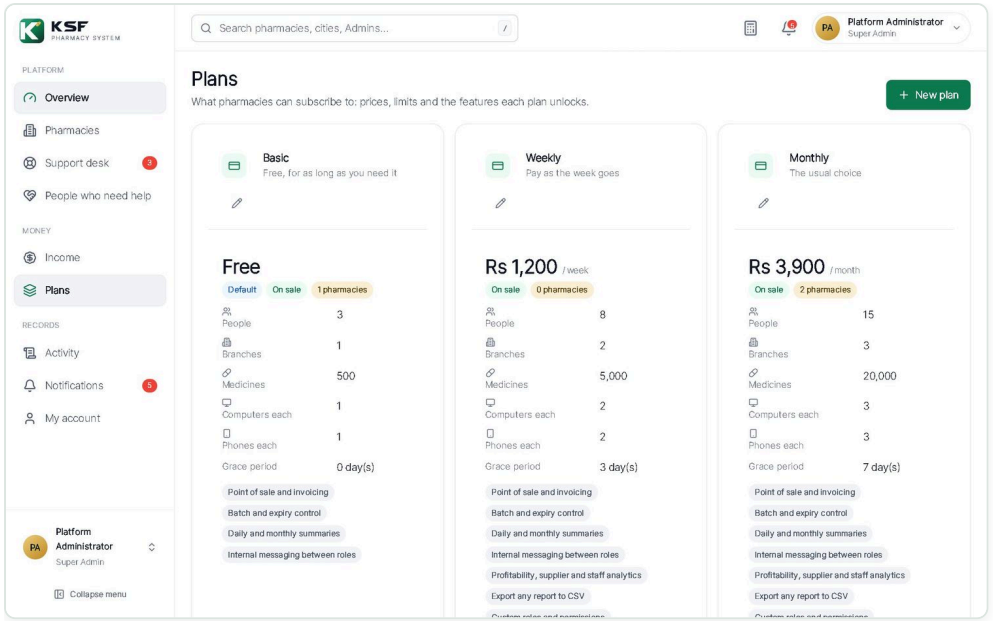


Anyone, in any pharmacy, who could not sign in today — locked out, wrong password, device limit — with a button to fix it. A search finds any account on the platform.

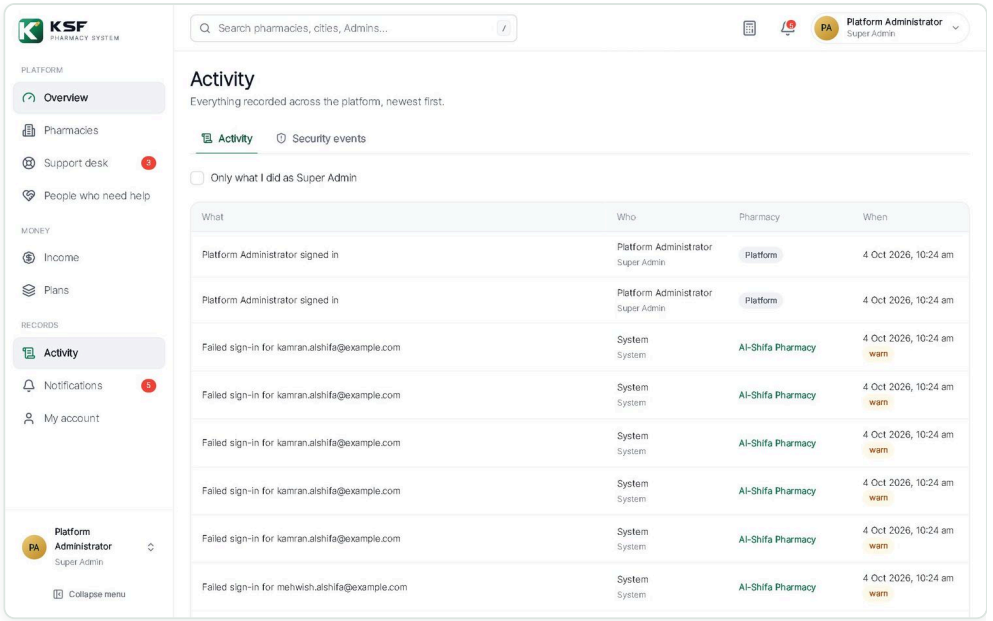
Income, plans and records



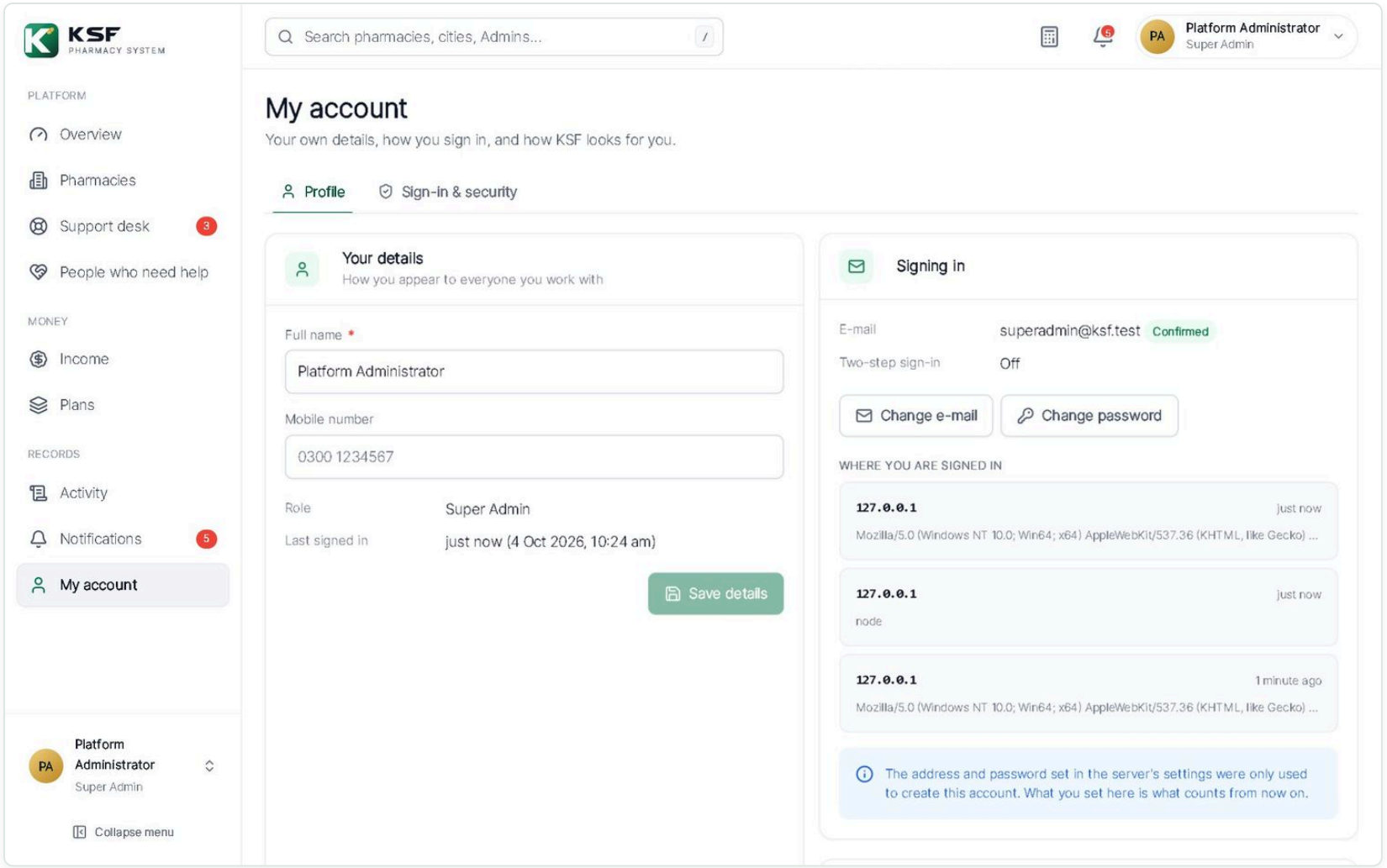
Income: payments by month, by plan and method, and by pharmacy; bank transfers waiting to be confirmed; refunds.



Plans: prices, limits (people, branches, medicines, computers, phones) and what each plan includes.



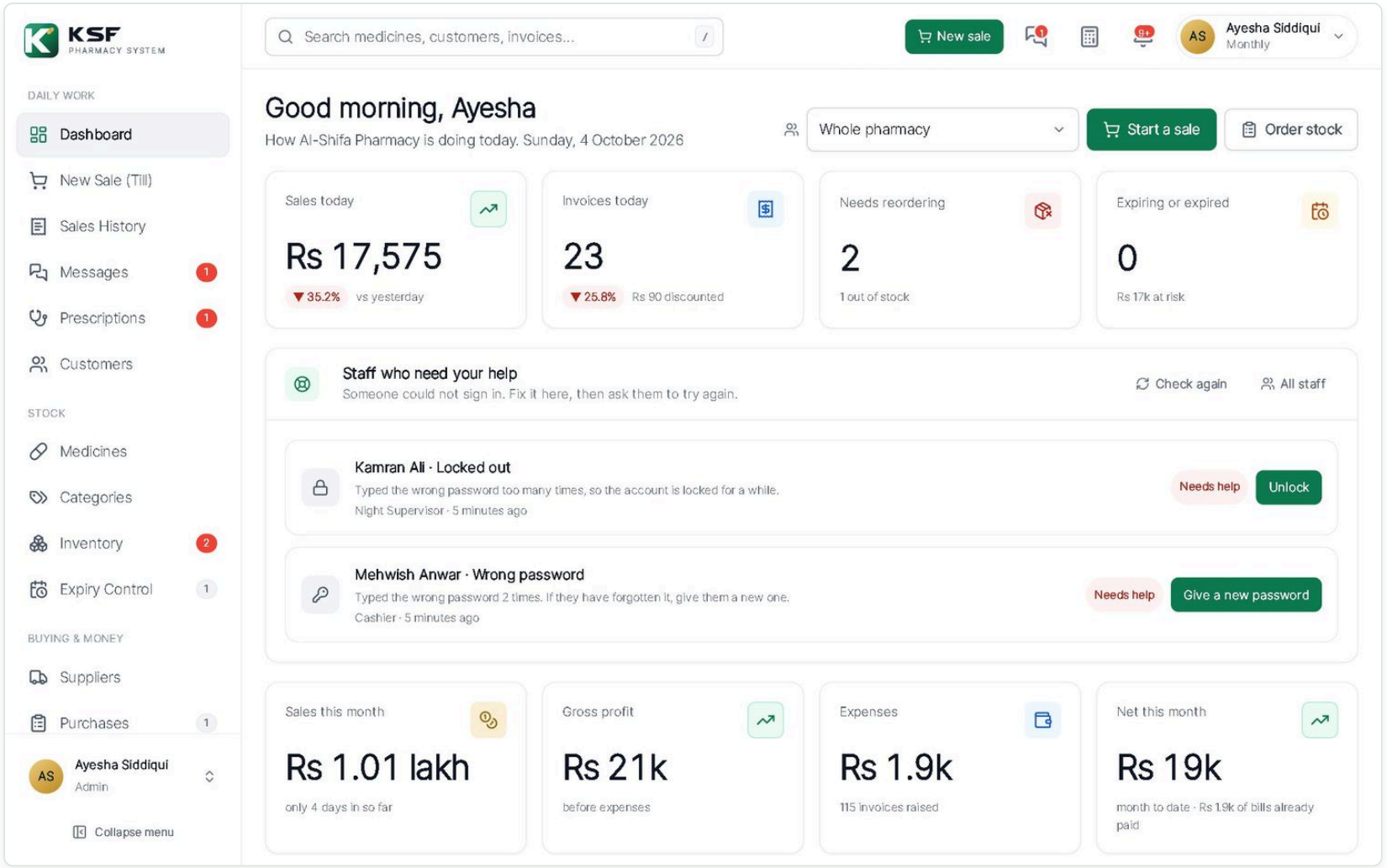
Activity: what happened across the platform, including every action taken in support mode.



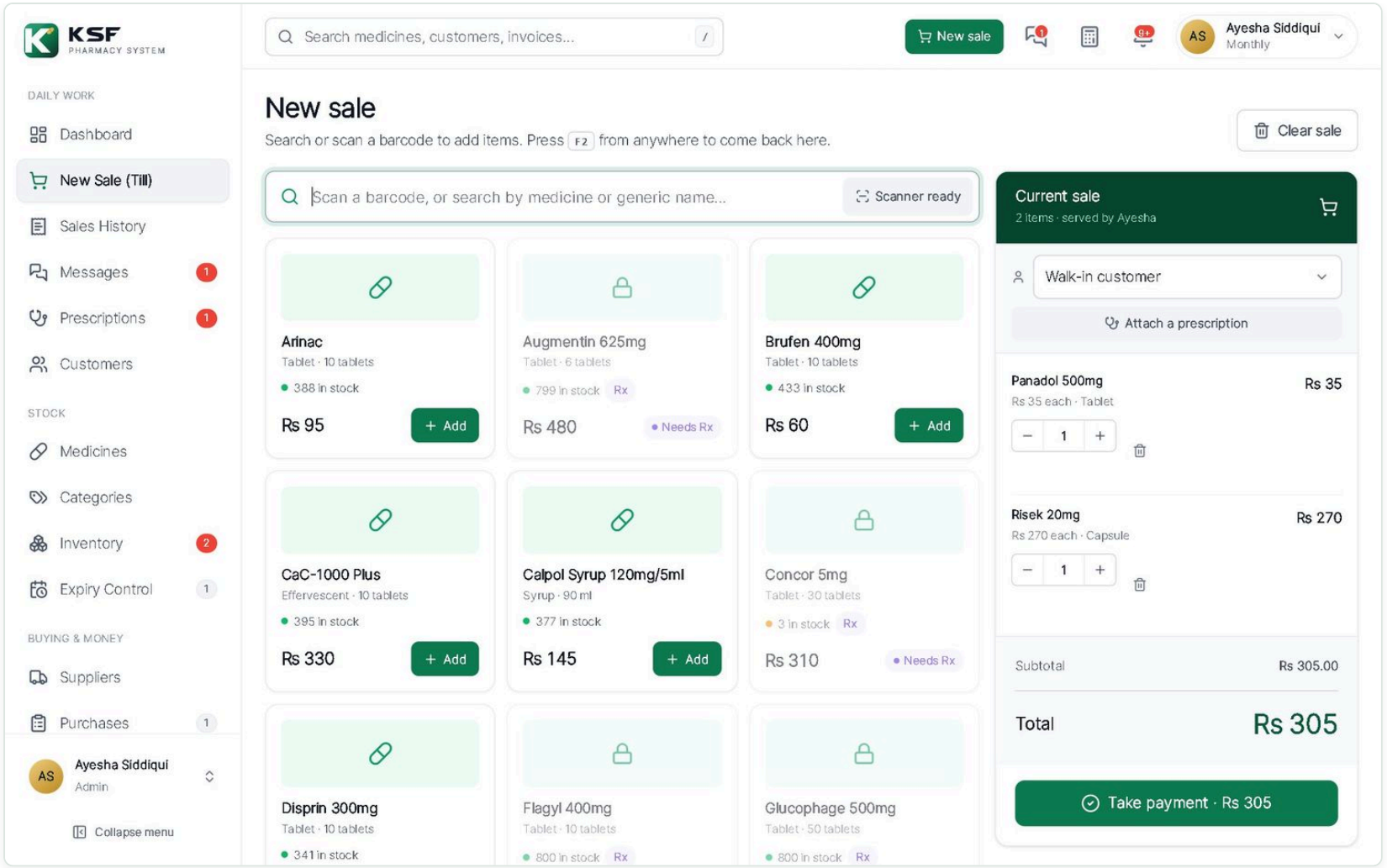
My account: the Super Admin changes his name, e-mail address and password here, and sees where he is signed in.

8 A pharmacy's screens

Everything a pharmacy used in 1.0.6 is still here and works the same way. These screens are from the sample pharmacy "Al-Shifa Pharmacy", signed in as its Admin.

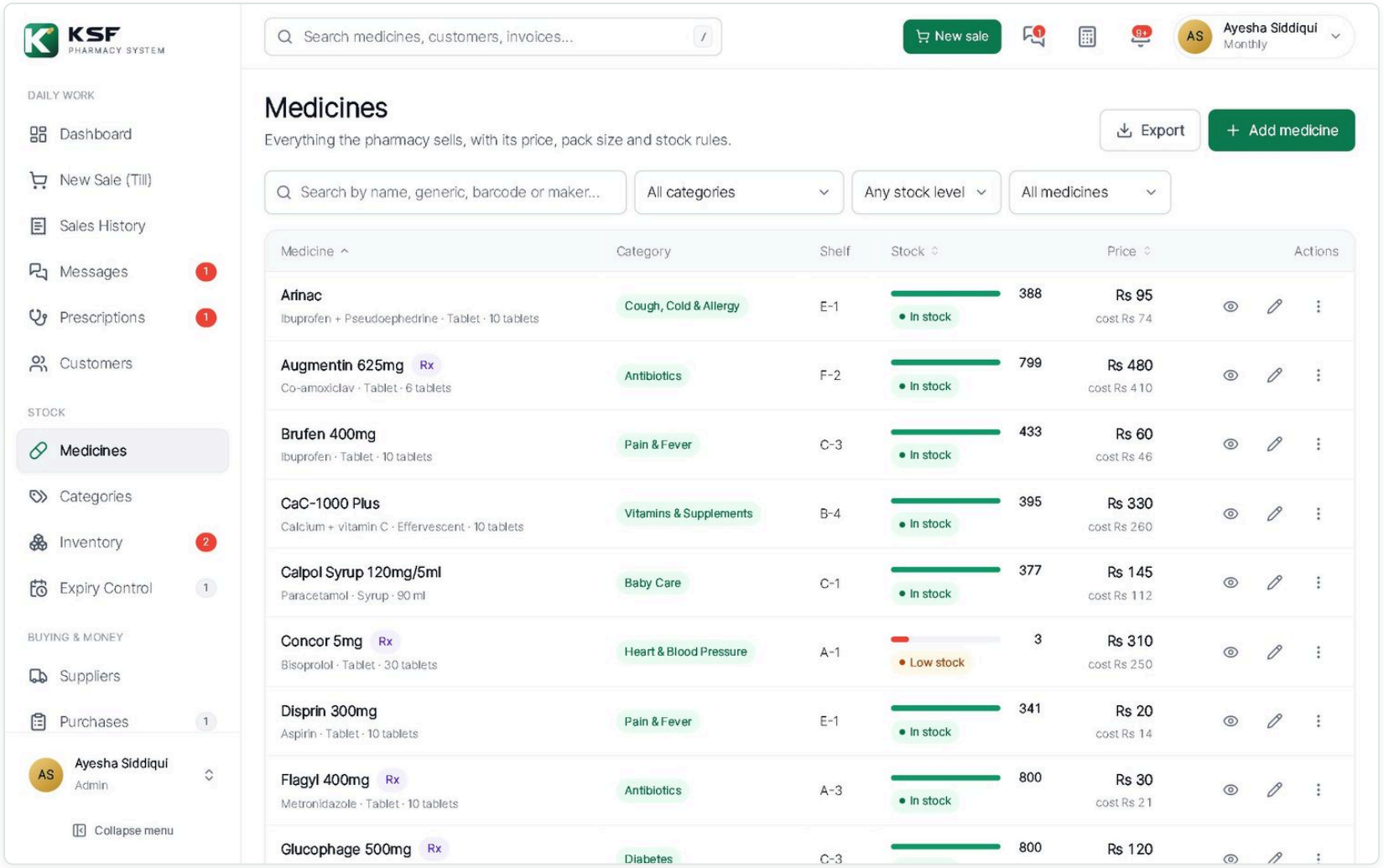


Dashboard: today's sales, what needs reordering or is expiring, staff who could not sign in, and the month so far.

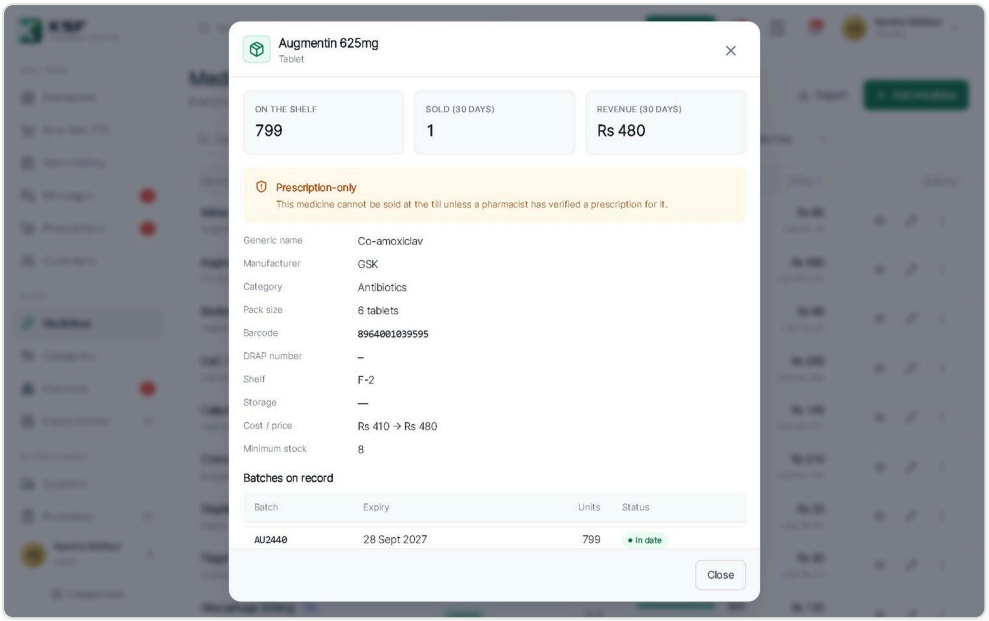


The till: search or scan, the batch nearest to expiry is used first, then take payment by cash, card, Easypaisa, JazzCash, bank transfer or credit (udhaar).

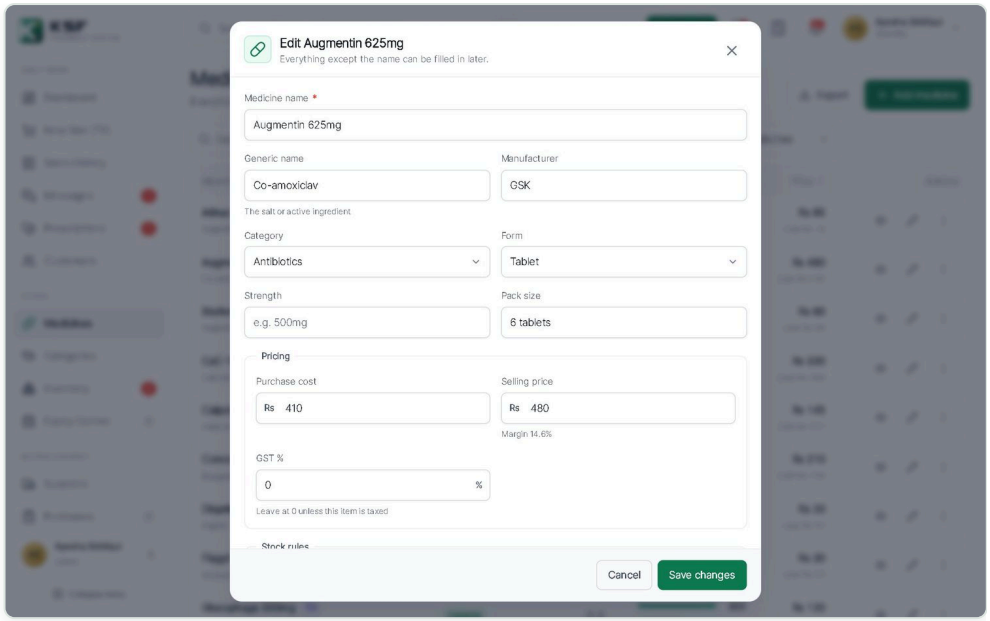
Medicines — view, edit and the row menu



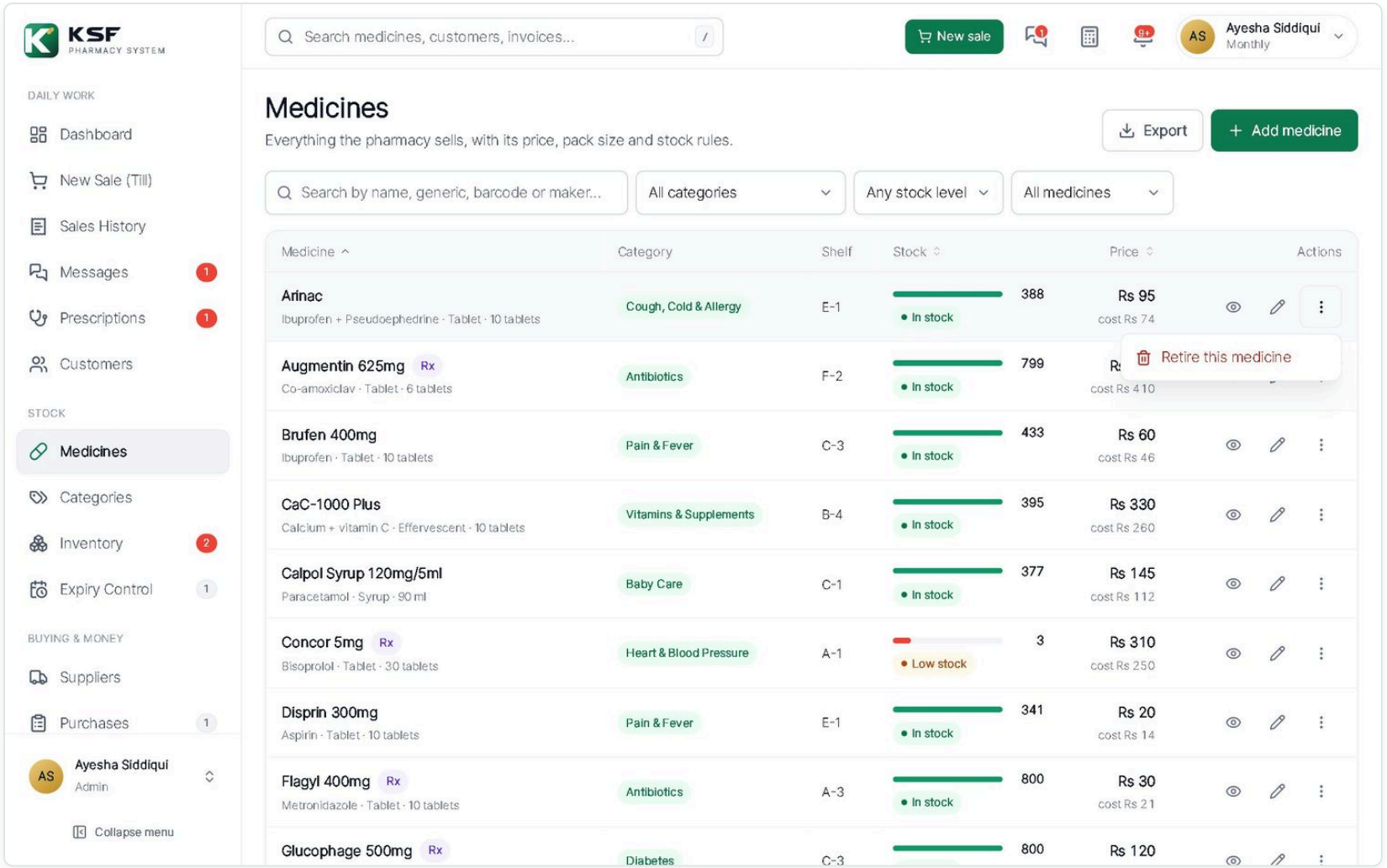
Medicines: price, pack, shelf and stock level at a glance.



View: everything about one medicine, with its batches.

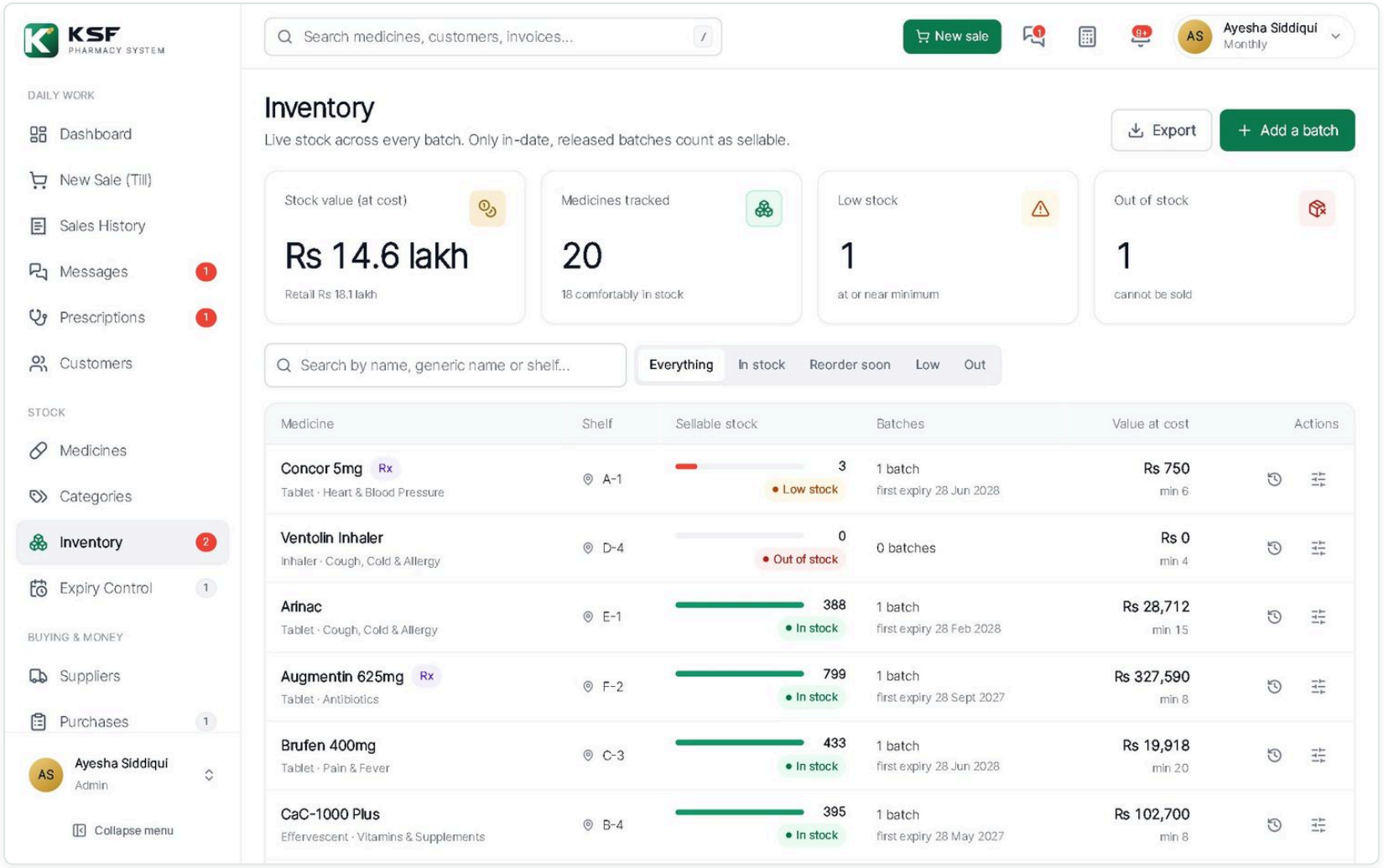


Edit: the same window used to add a medicine.

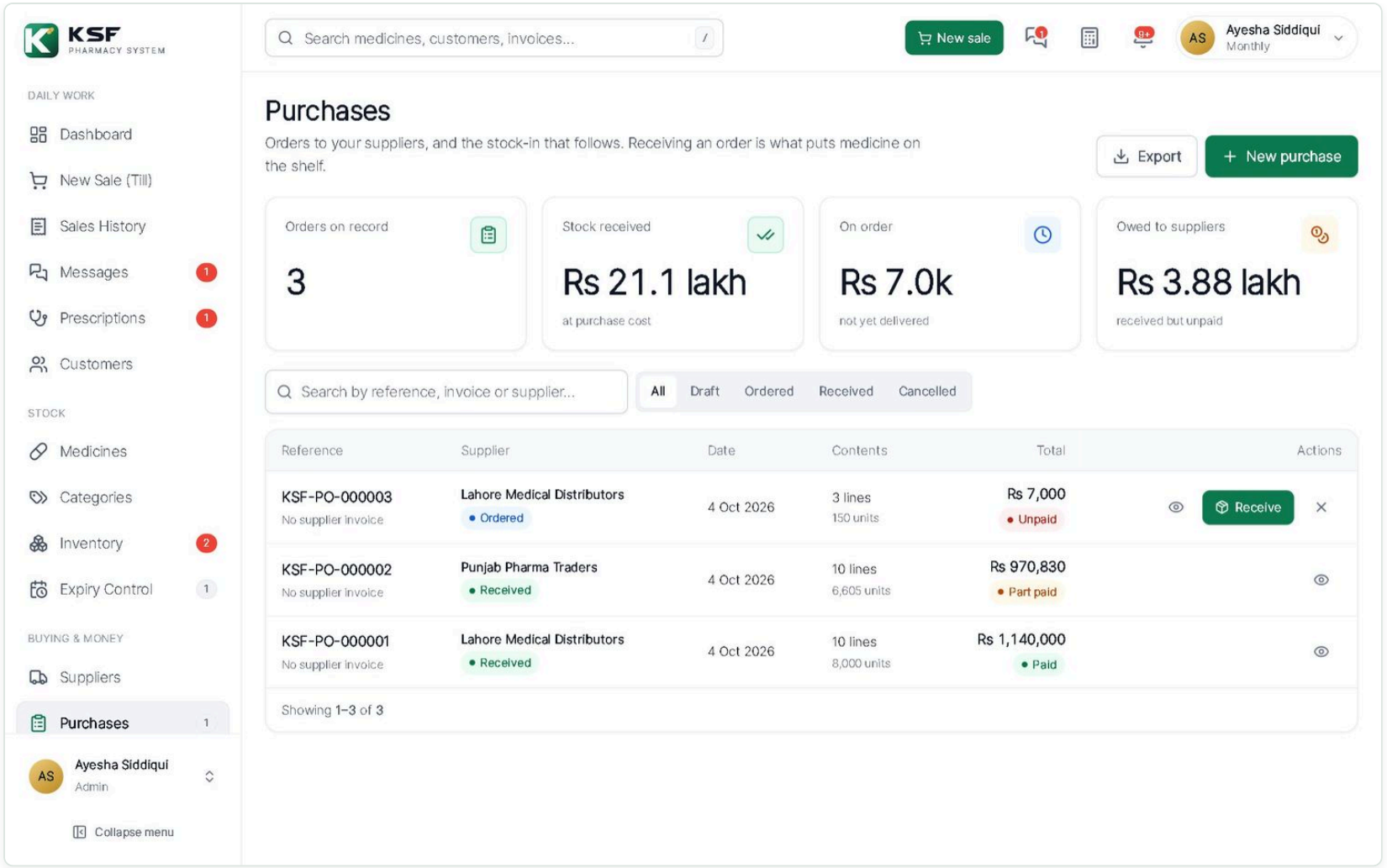


The "more actions" menu now opens on top of the table instead of being cut off by it.

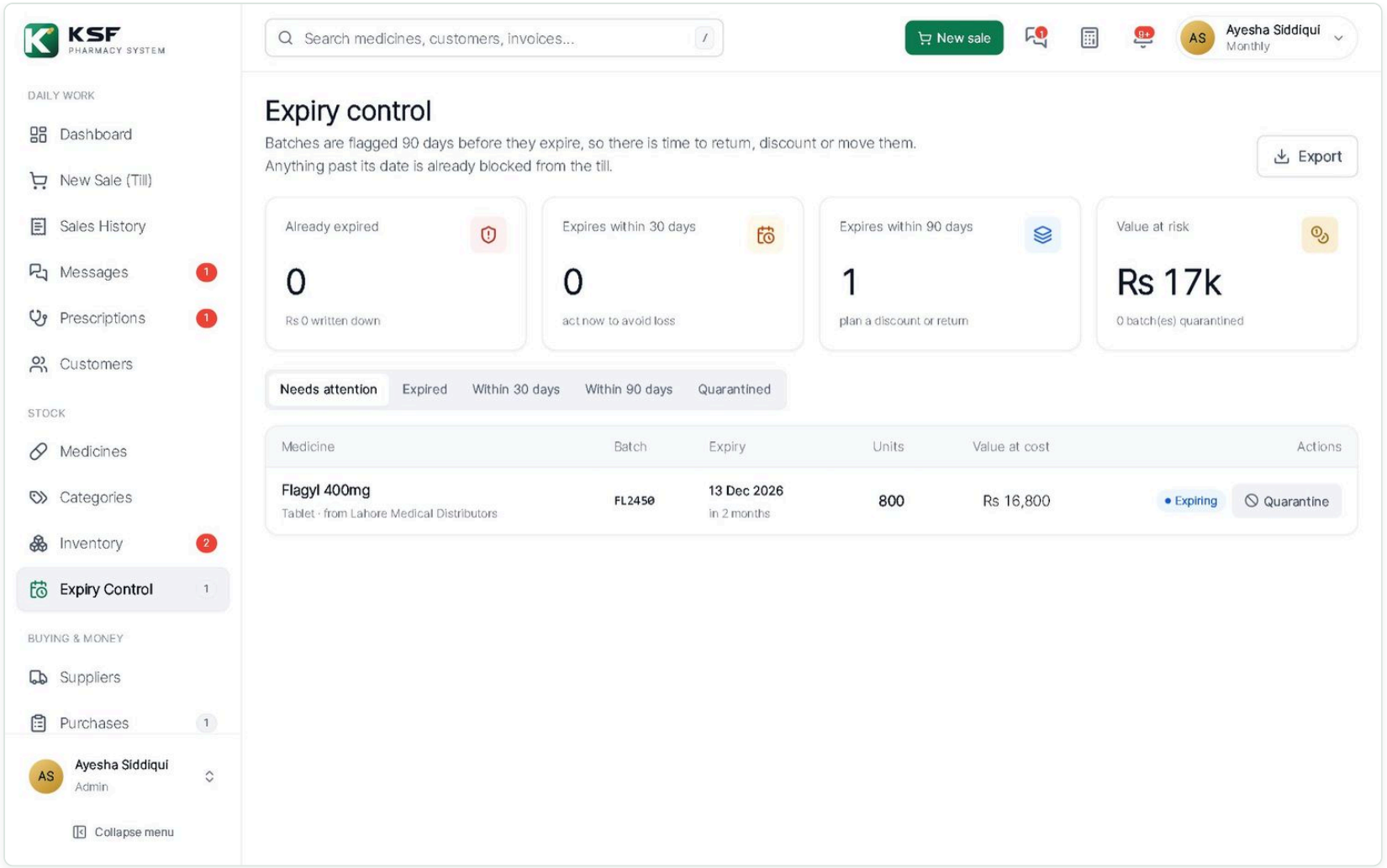
Stock and buying



Inventory: stock value, low and out-of-stock items, every batch.

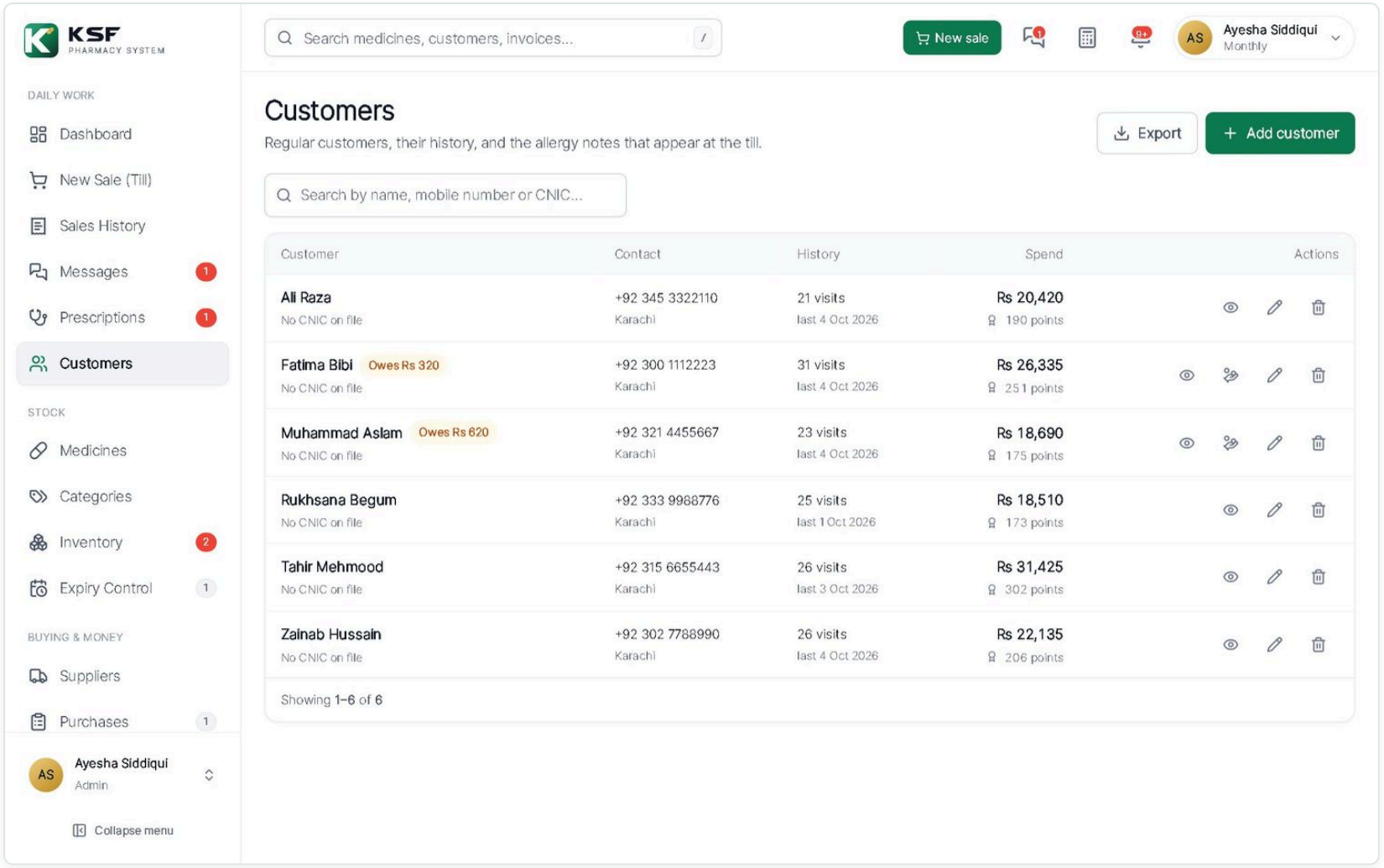


Purchases: orders to suppliers, receiving into stock, and what is still owed.

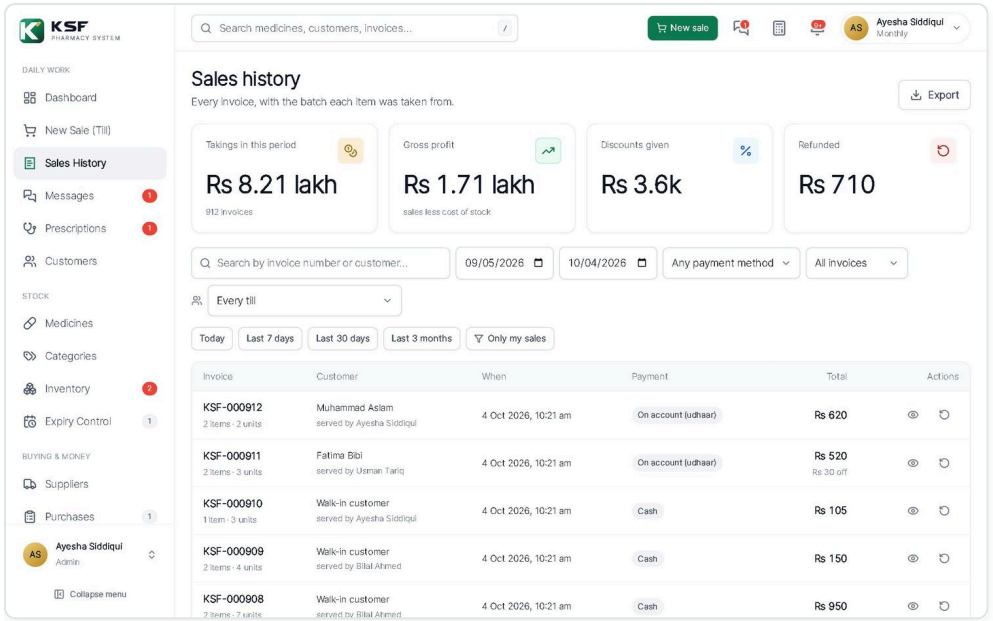


Expiry control: what expires soon, and quarantine or write-off for batches that should not be sold.

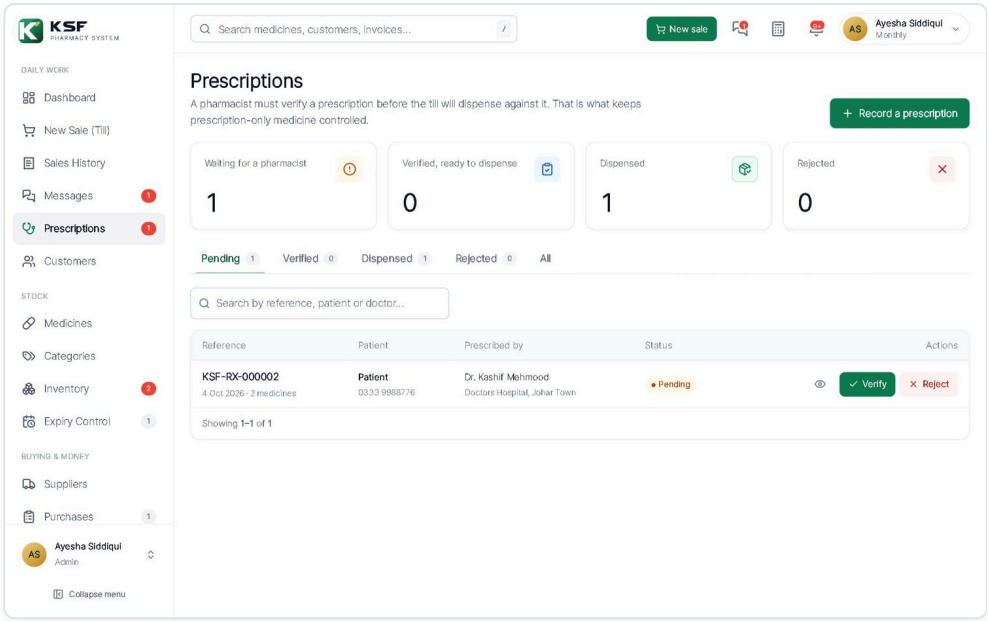
Customers, sales and prescriptions



Customers: visits, spending, credit (udhaar) owed and allergies.

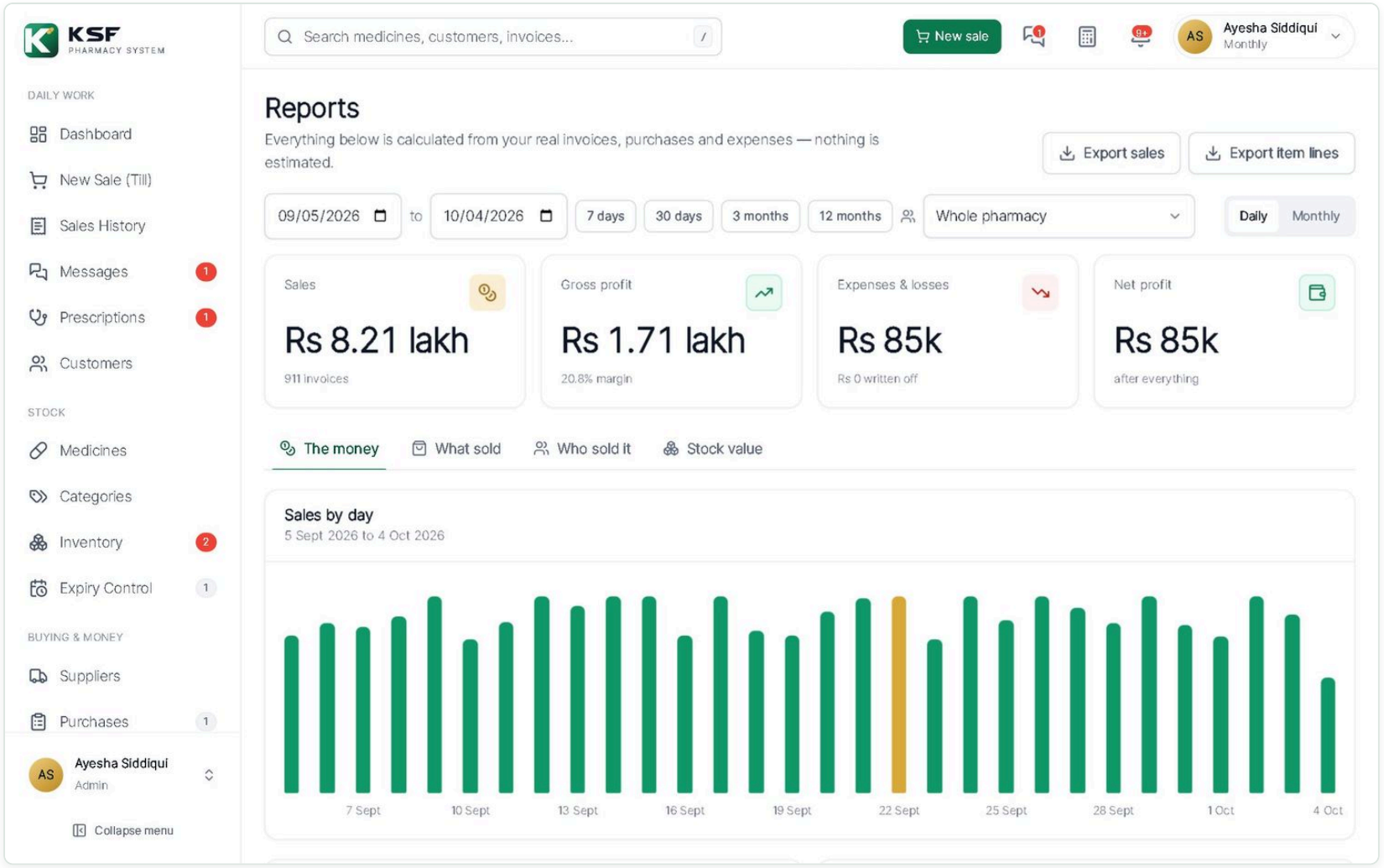


Sales history: every invoice, with filters by date, payment method and person.

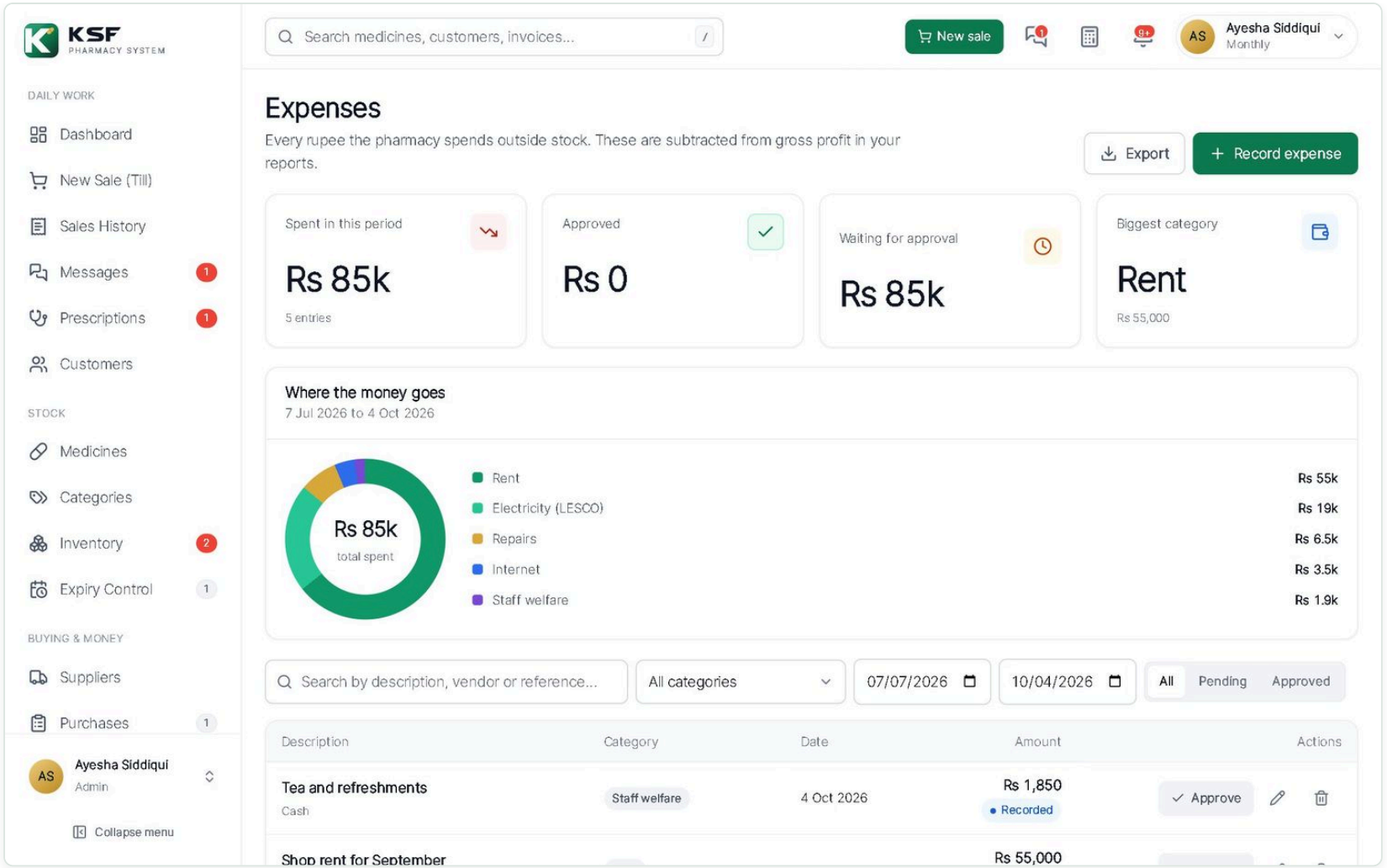


Prescriptions: waiting for a pharmacist, verified, dispensed.

Money and reports

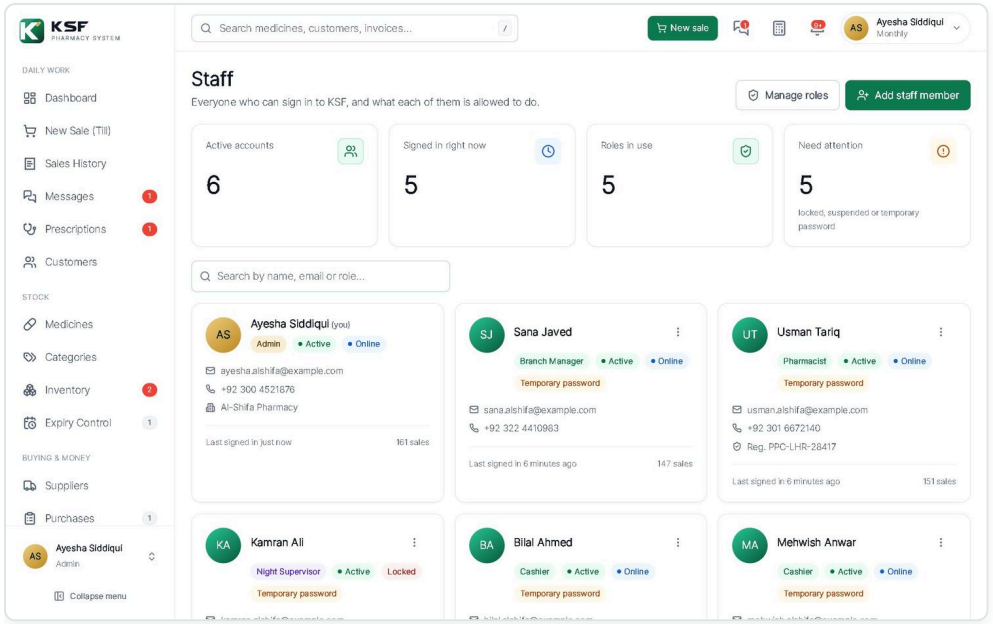


Reports: sales, gross profit, expenses and net profit for any period, by day or by month; what sold, who sold it and stock value.

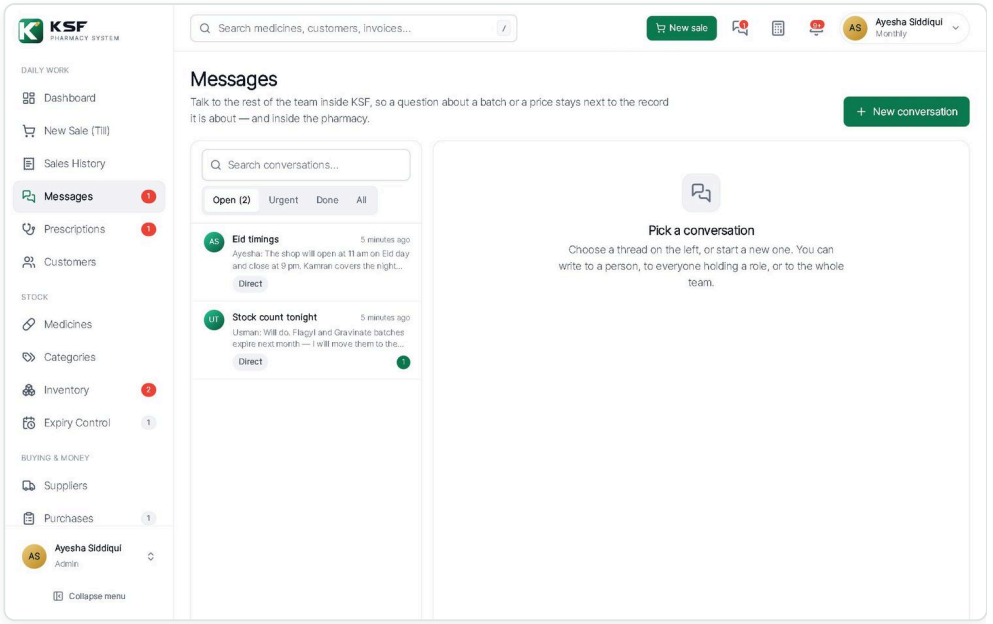


Expenses: what the pharmacy spends outside stock, by category.

Team

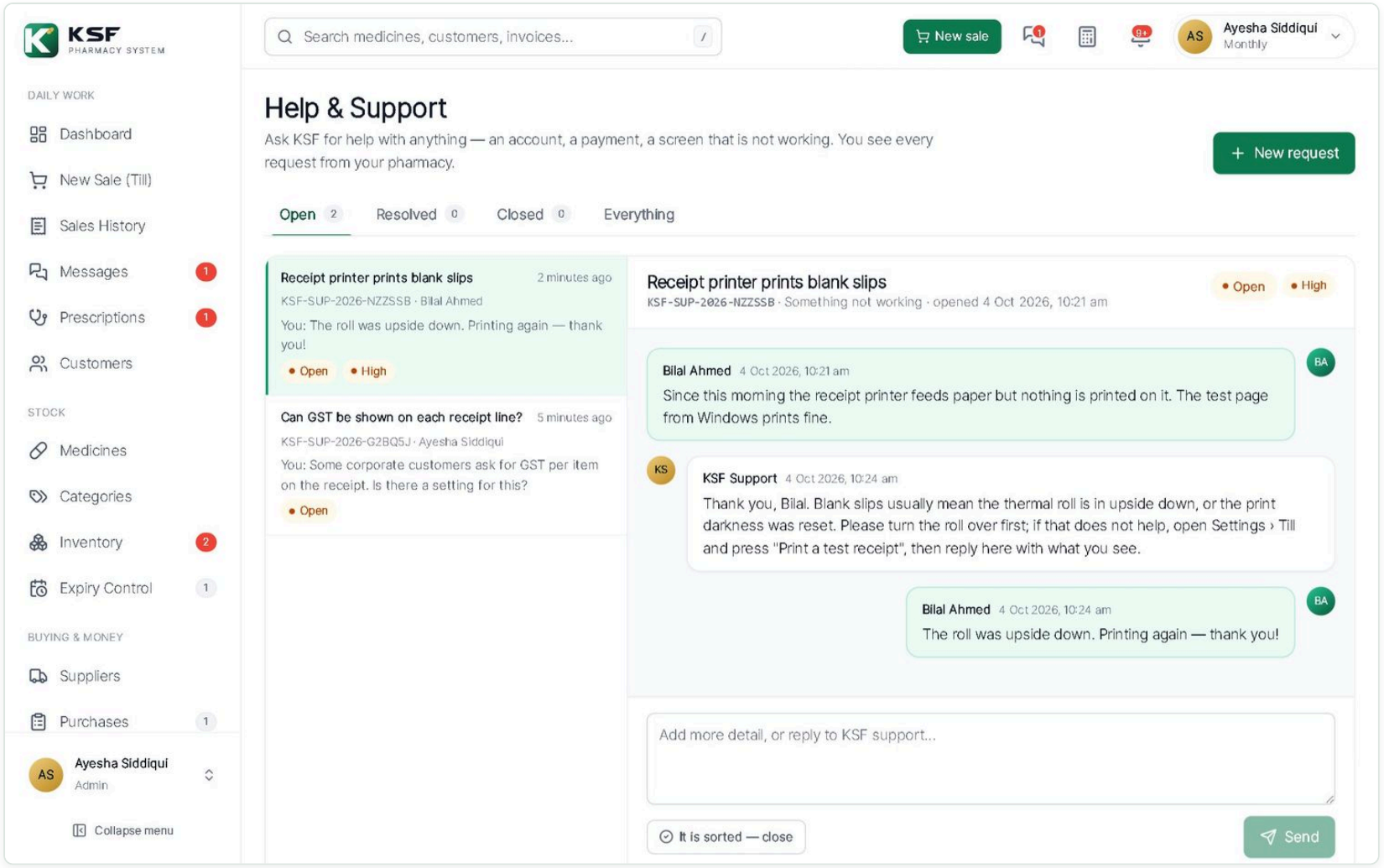


Staff: who works here, their role, and who needs attention (for example, locked out).

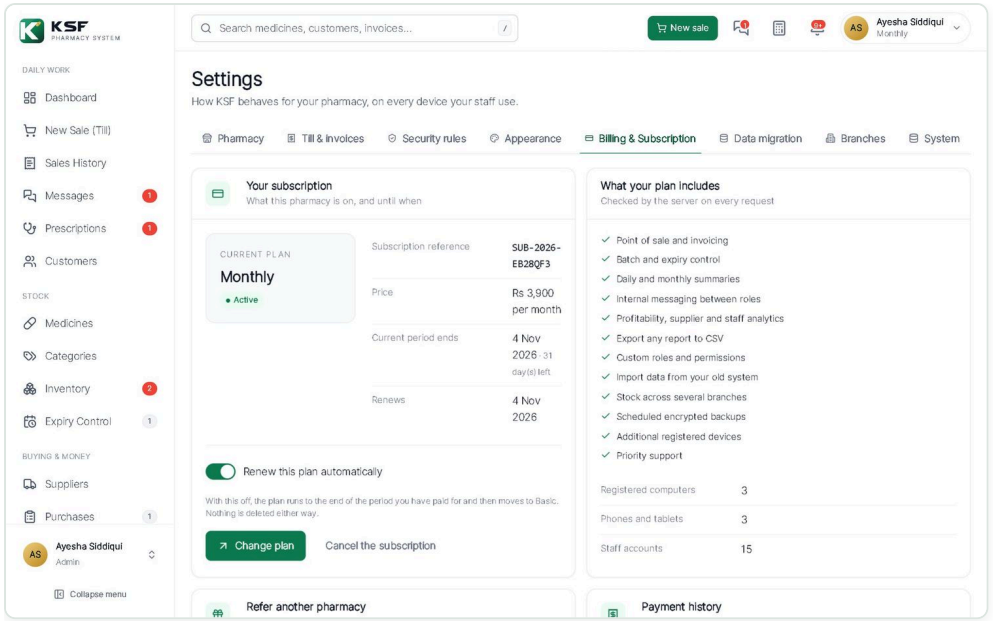


Messages between staff: to one person, to a role, or to the whole team.

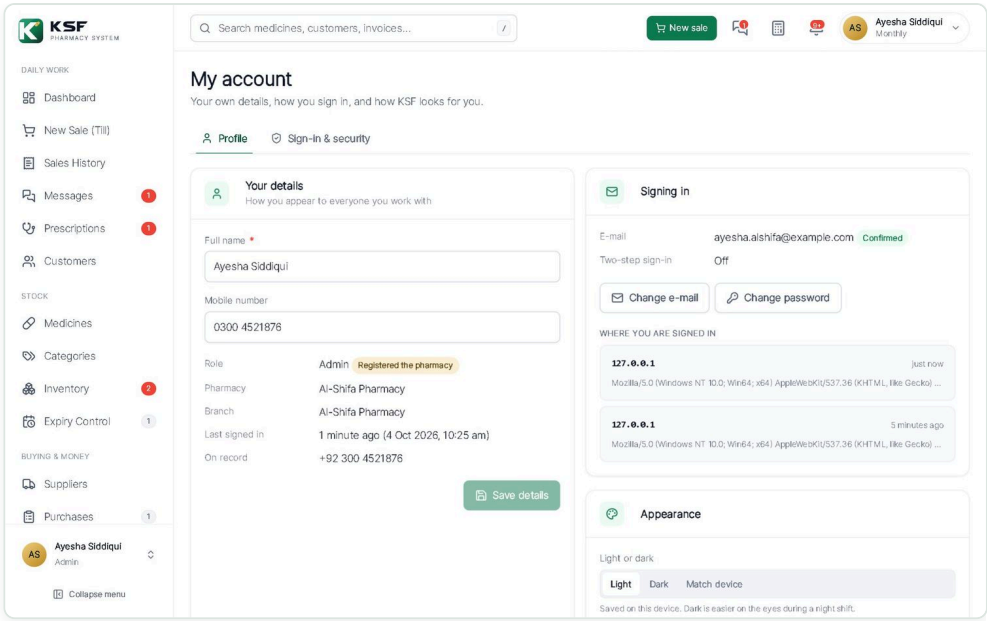
Help & Support, billing and My account



Help & Support (new): a pharmacy raises a request to KSF and reads the Super Admin's reply here.



Billing: the current plan, what it includes, payments and referrals.



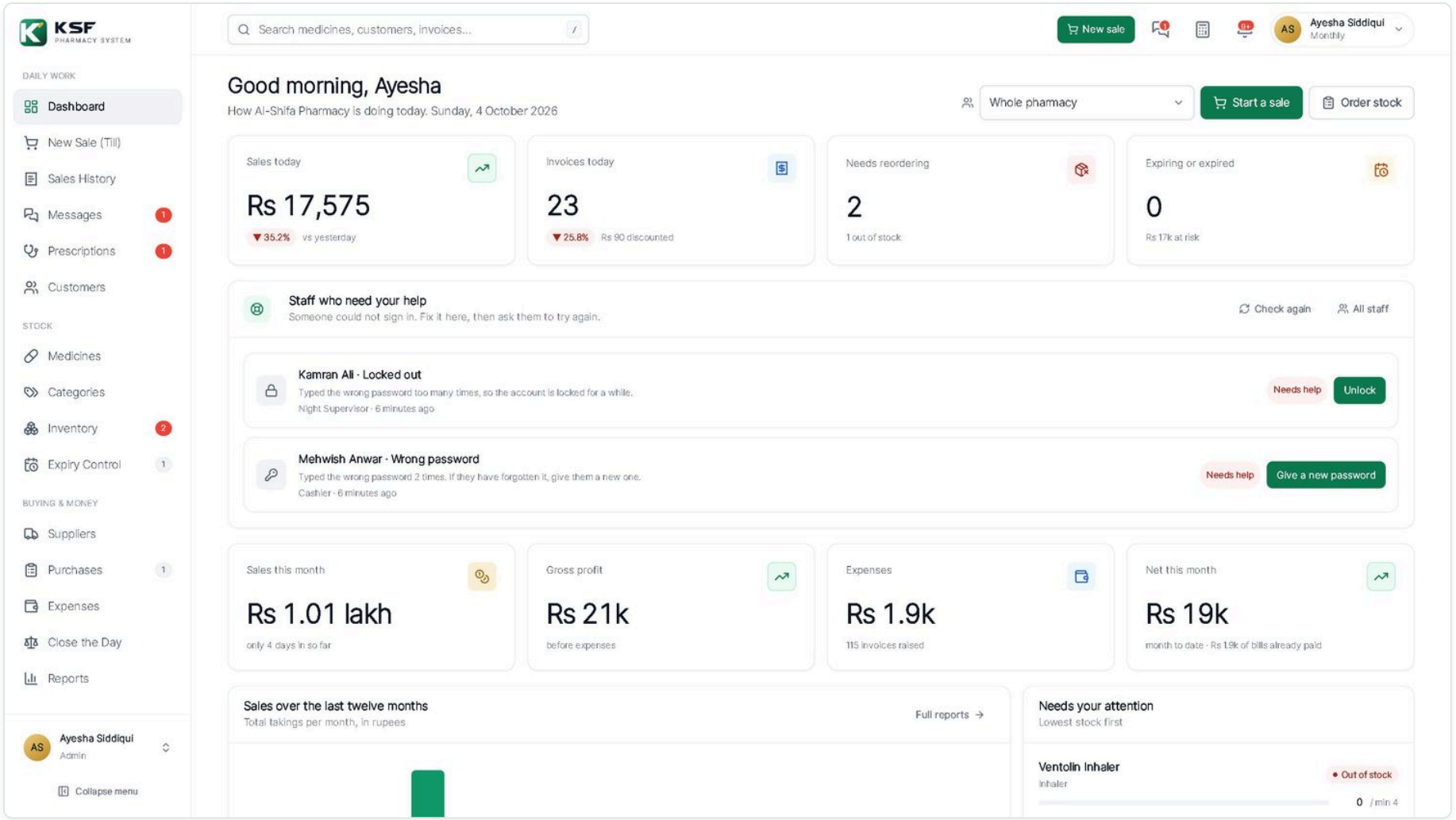
My account: each person's own details, password, two-step sign-in and devices, in one place.

9 Full width, and every screen size

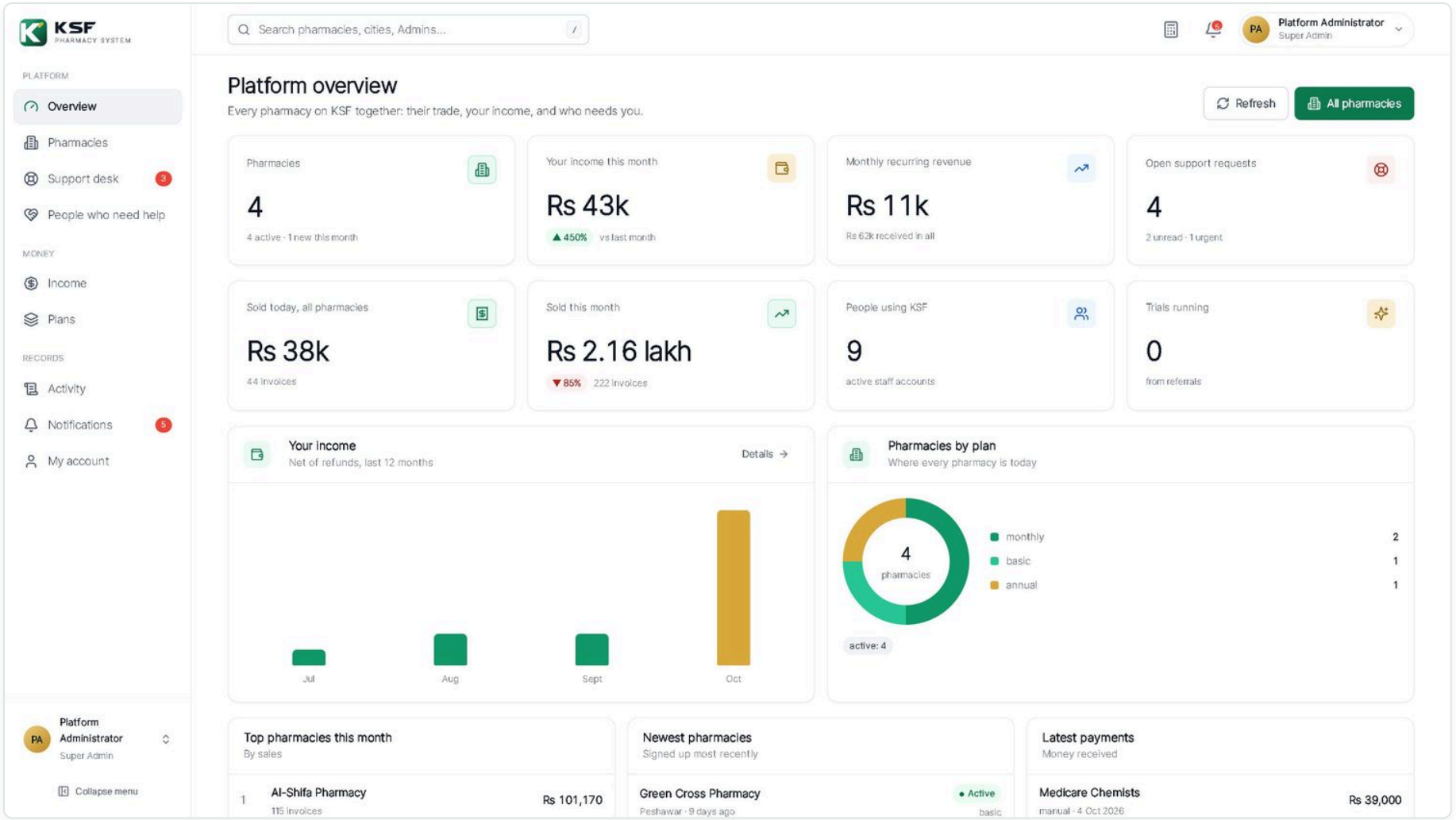
The app now uses the whole screen on a laptop or a wide monitor, with no empty margins around it, and rearranges itself on tablets and phones.

Size	Width	What happens
Phones	under 640 px	Cards stack; the menu becomes a drawer behind the ☰ button; the five daily actions sit in a bar at the bottom
Tablets	640 – 1099 px	Two columns of cards; the menu is still a drawer
Laptops	1100 – 1535 px	Menu on the left, four cards across
Wide monitors	1536 px and up	Uses the full width — no boxed-in page

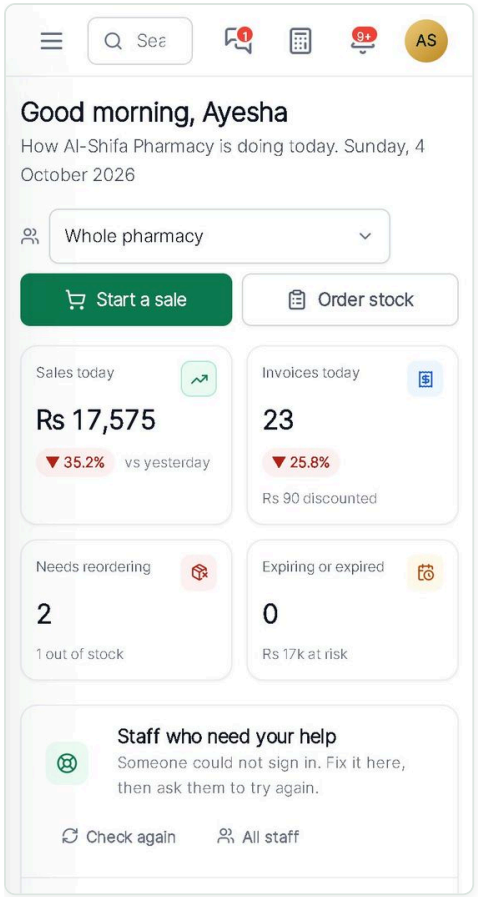
Behind this are seven named sizes used the same way in every screen: under 360, 360, 480, 640, 1024, 1280 and 1536 pixels.



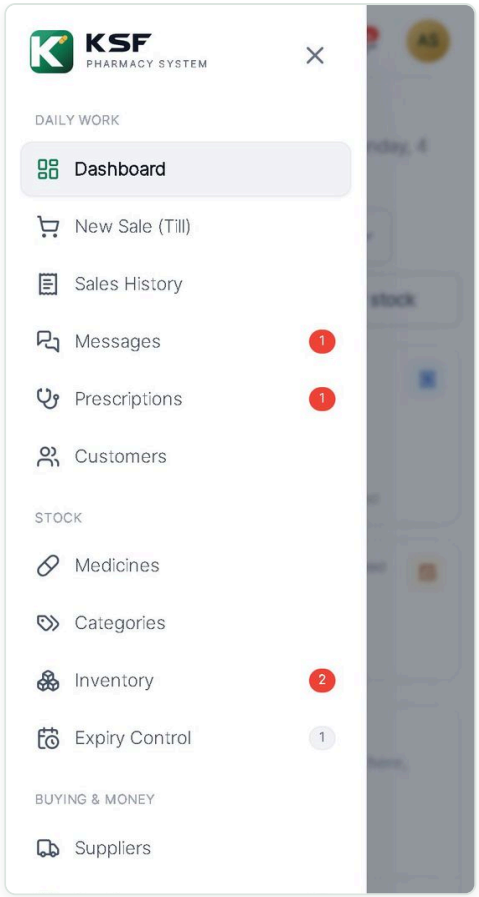
A wide monitor (1920 px): the page fills the screen.



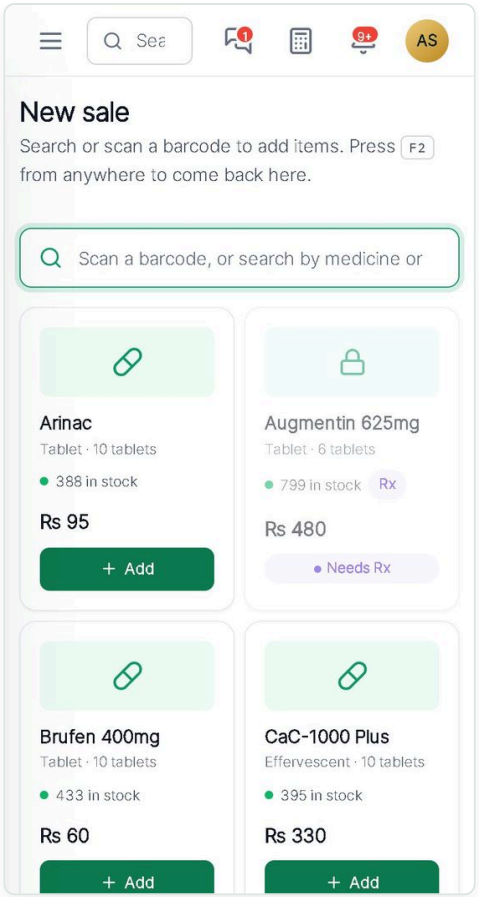
The Super Admin's overview on the same monitor.



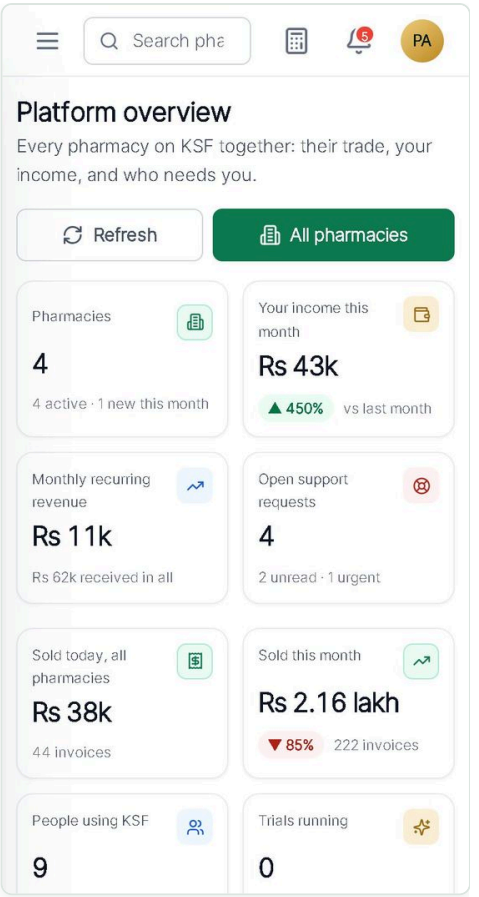
Phone: dashboard



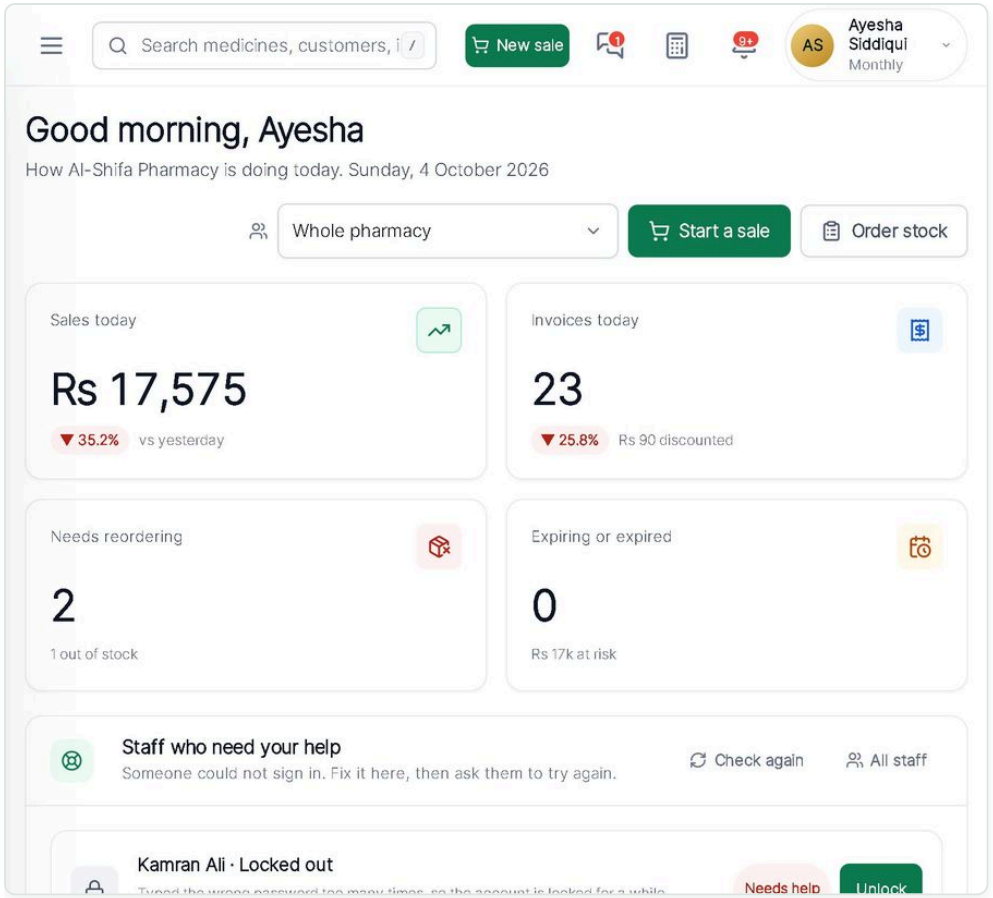
Phone: the menu



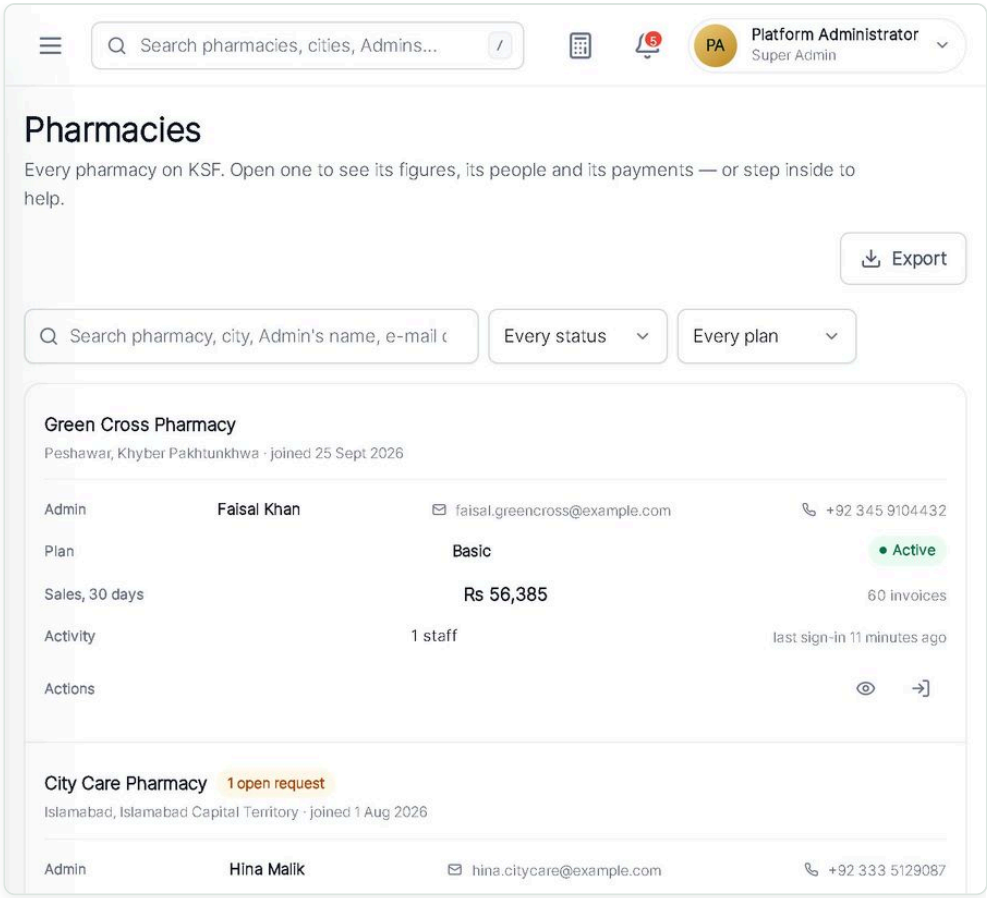
Phone: the till



Phone: Super Admin



Tablet: dashboard

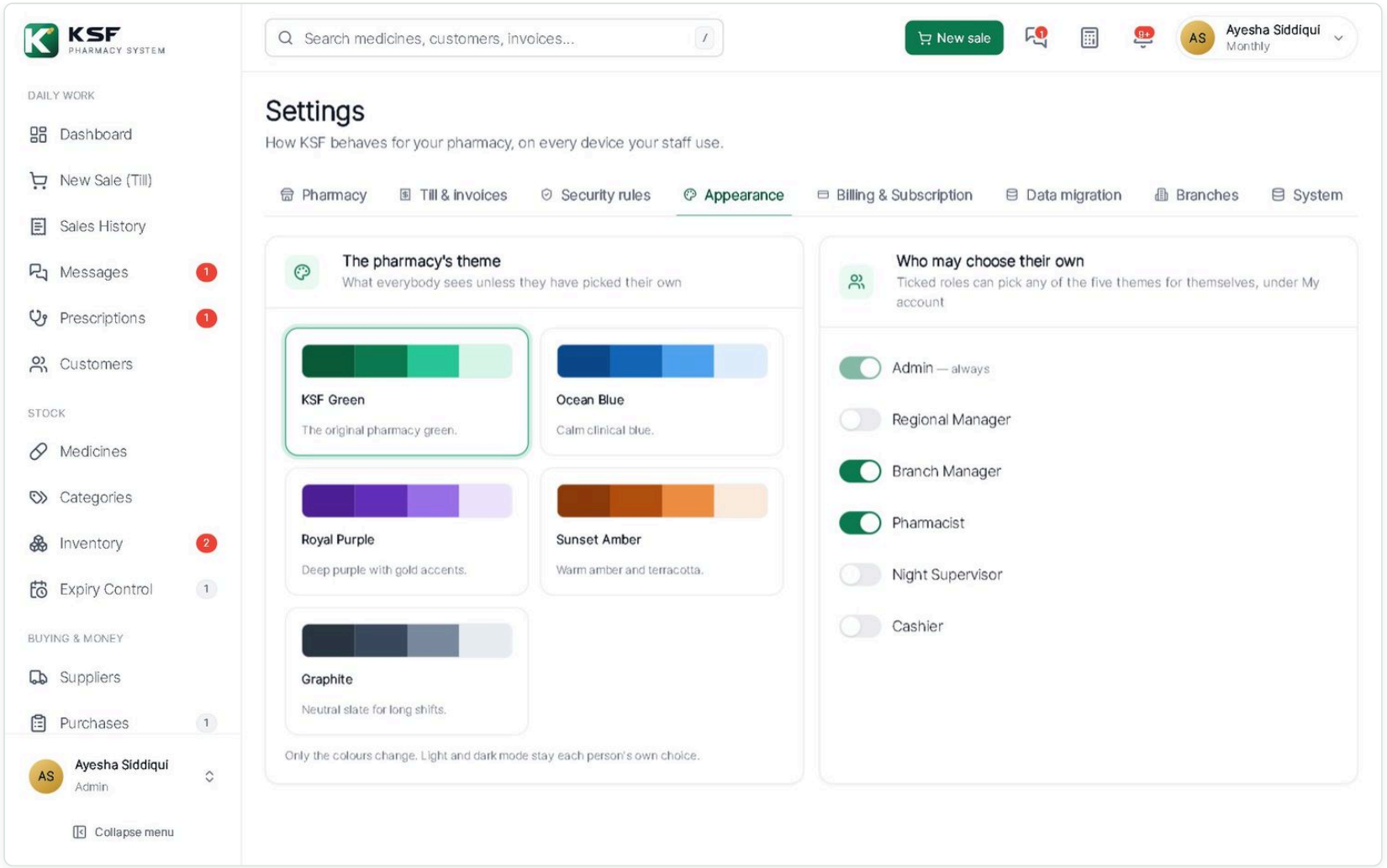


Tablet: Super Admin's pharmacies

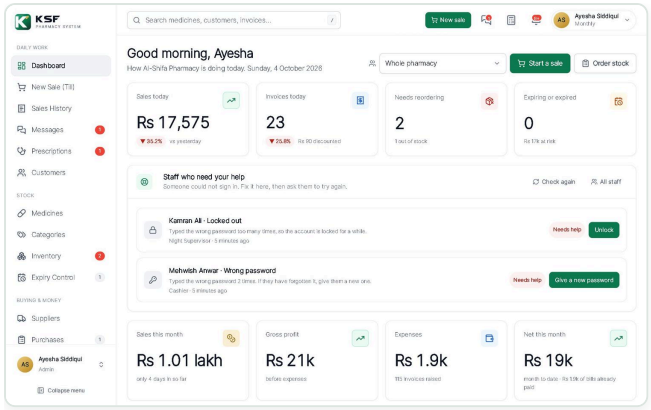
10

Five themes

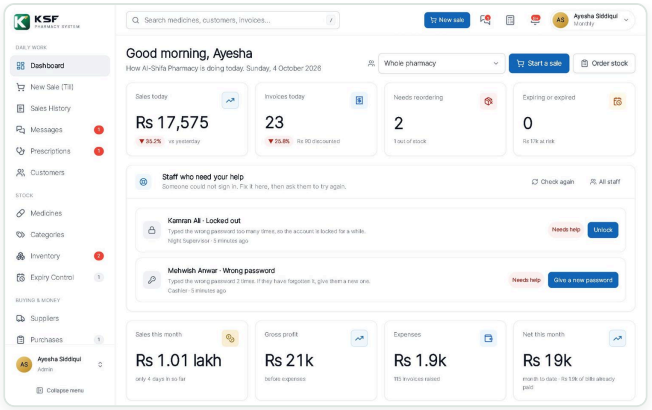
KSF Green stays the default. The pharmacy's Admin picks the pharmacy's theme and decides which roles may choose their own. Every theme has a light and a dark mode, and only the colours change — never the layout.



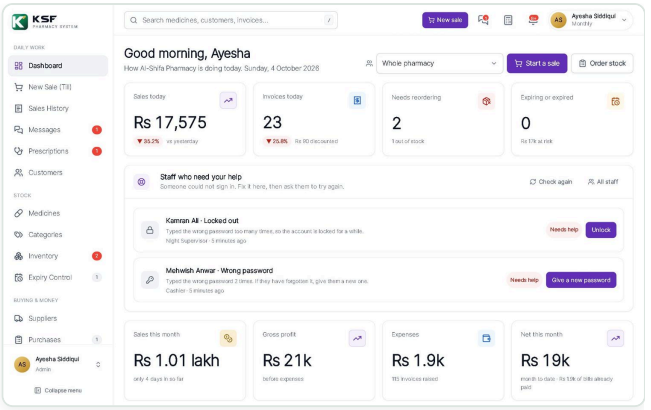
Settings › Appearance: the pharmacy's theme on the left; on the right, which roles may pick their own (here, Branch Manager and Pharmacist).



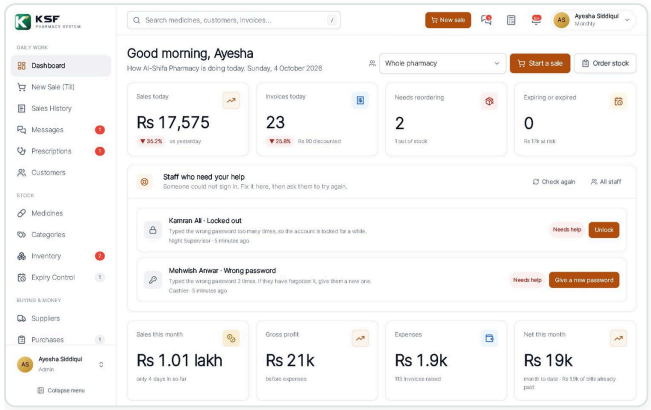
KSF Green (default)



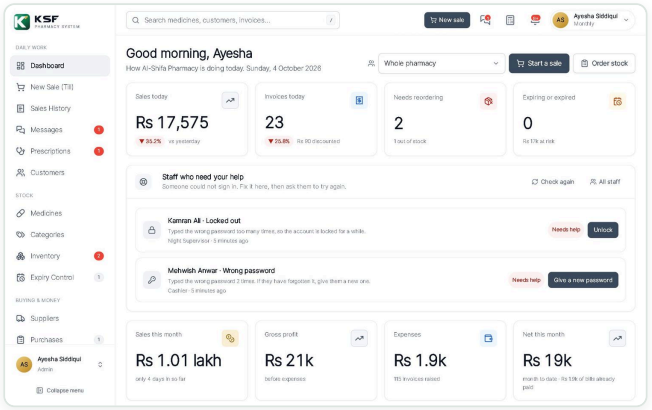
Ocean Blue



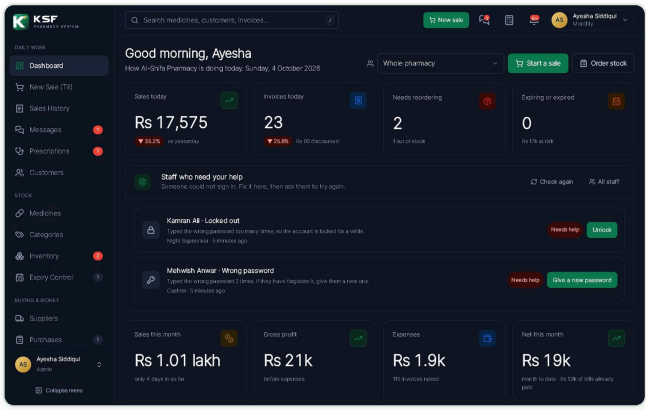
Royal Purple



Sunset Amber



Graphite



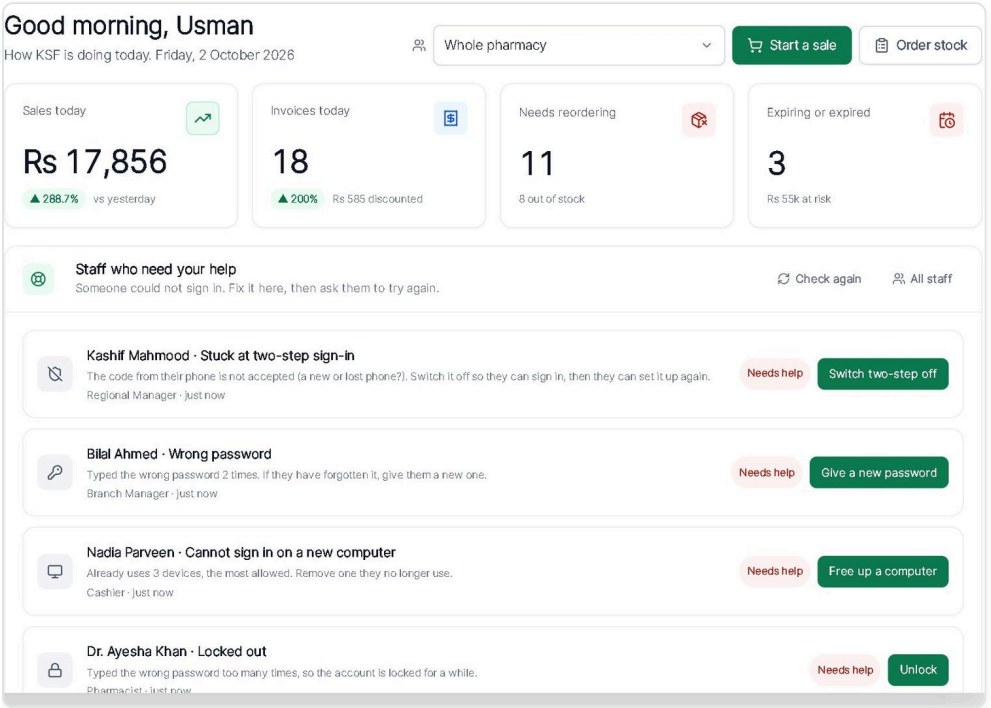
Dark mode (KSF Green)

11 Previous and new, side by side

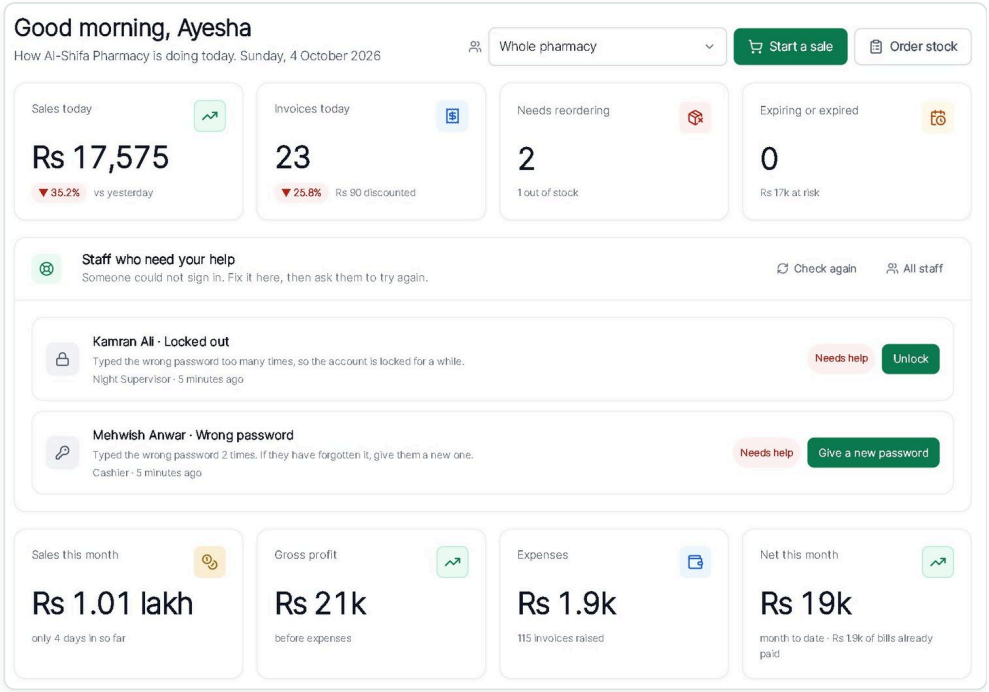
The same screens in version 1.0.6 (with its demonstration pharmacy, 2 October) and in version 2.0 (with the sample pharmacy). The design is the same on purpose; the figures differ because the data differs.

Dashboard

PREVIOUS · 1.0.6



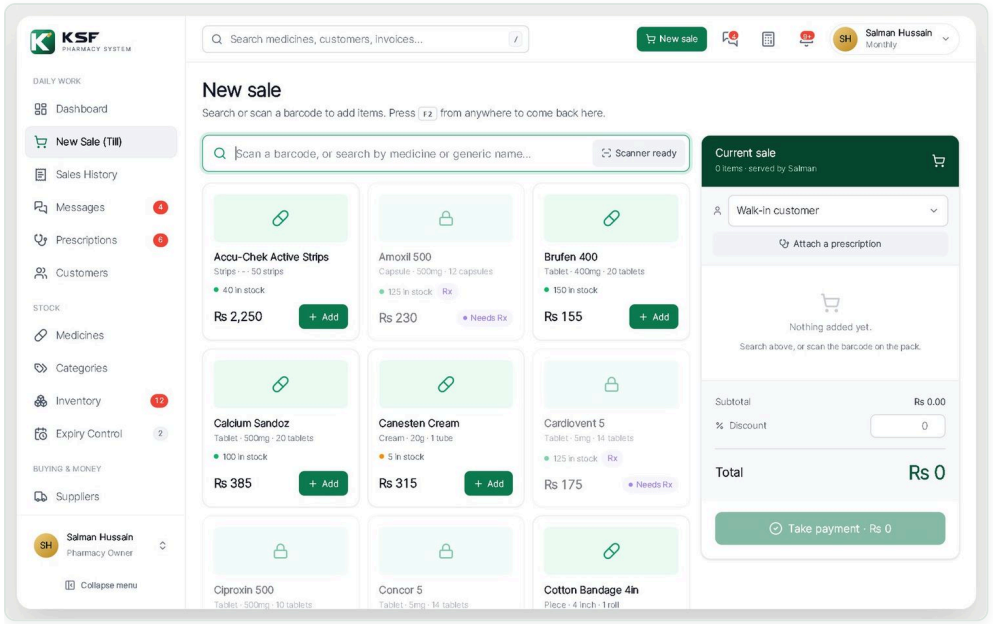
NEW · 2.0



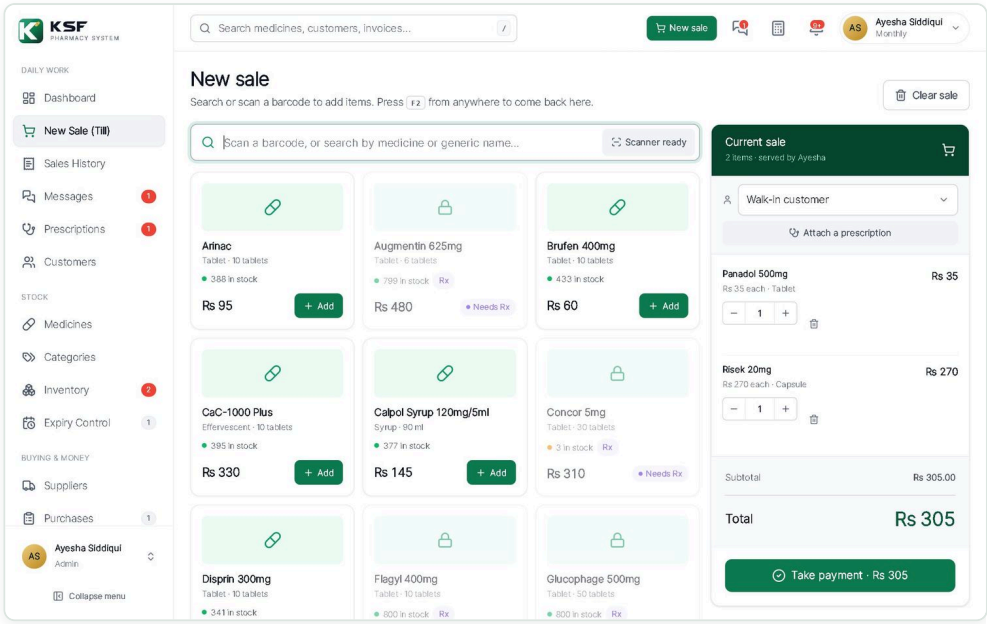
Same layout and the same cards; the figures come from different sample data.

The till

PREVIOUS · 1.0.6



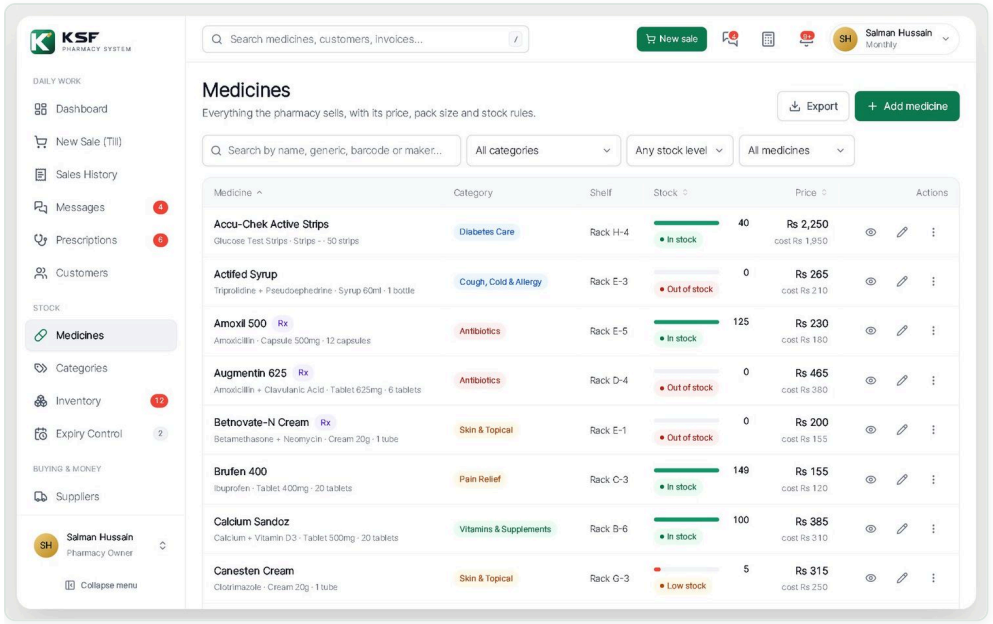
NEW · 2.0



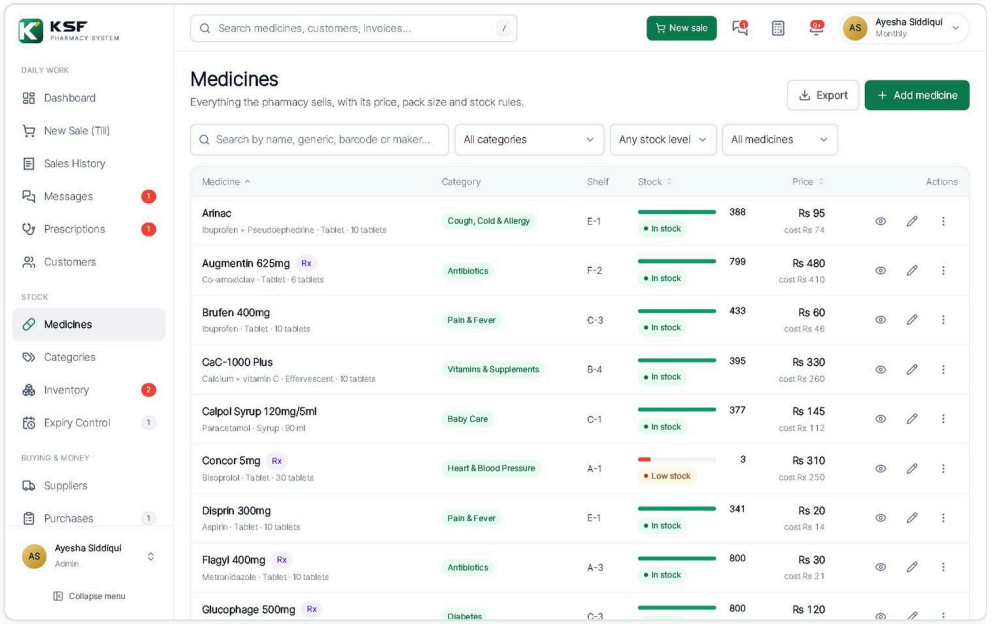
Same till and payment flow.

Medicines

PREVIOUS · 1.0.6



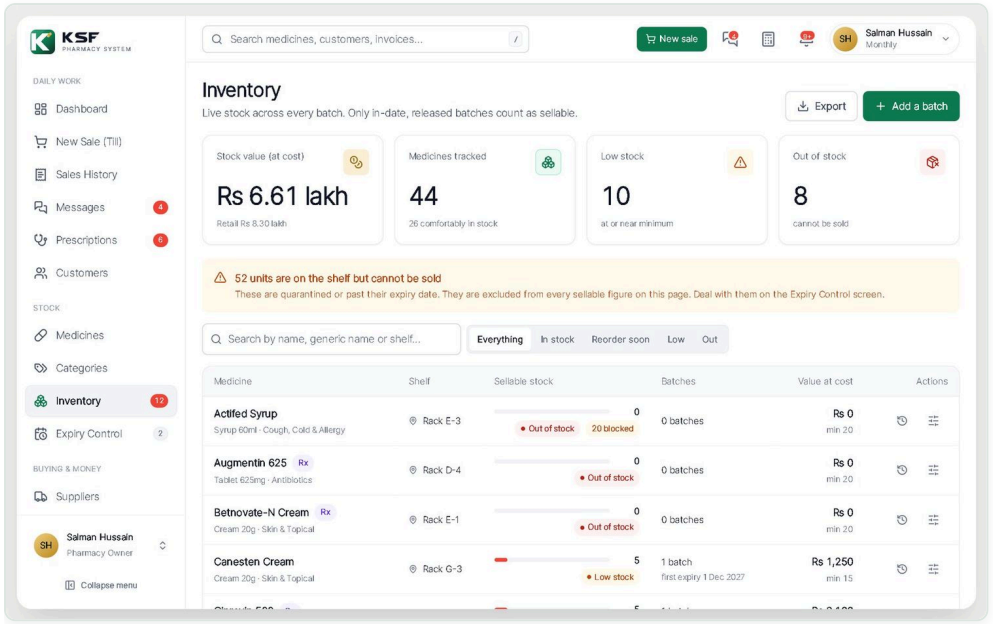
NEW · 2.0



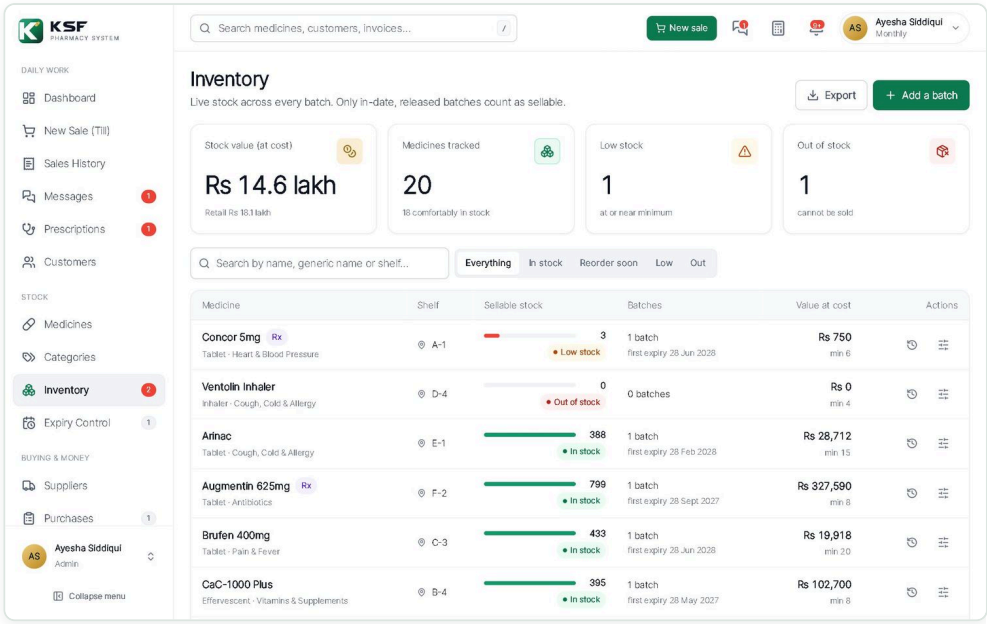
Same list; the row menu now floats above the table.

Inventory

PREVIOUS · 1.0.6



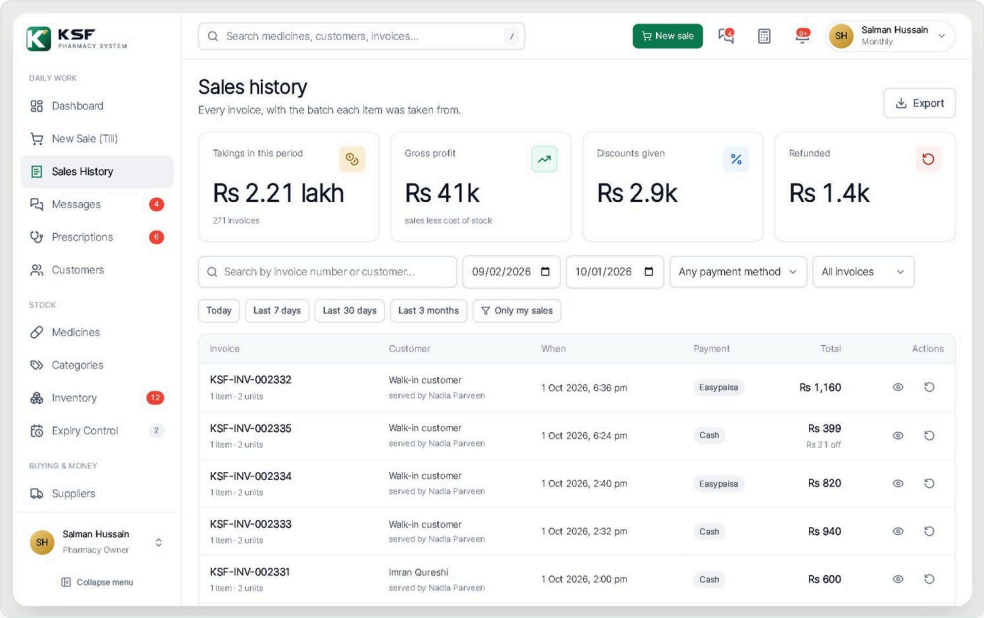
NEW · 2.0



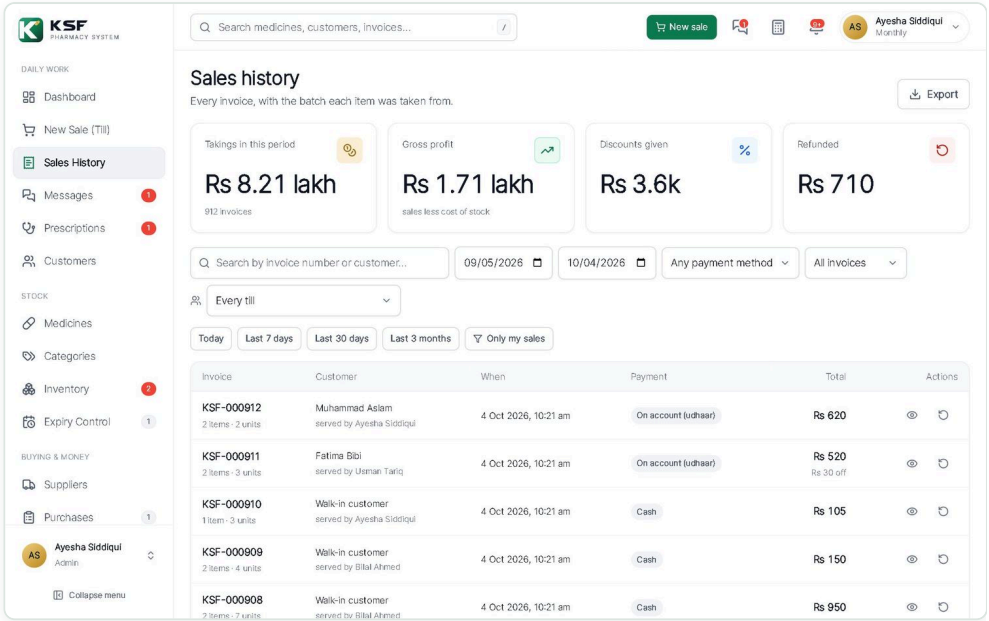
Same screen and figures.

Sales history

PREVIOUS · 1.0.6



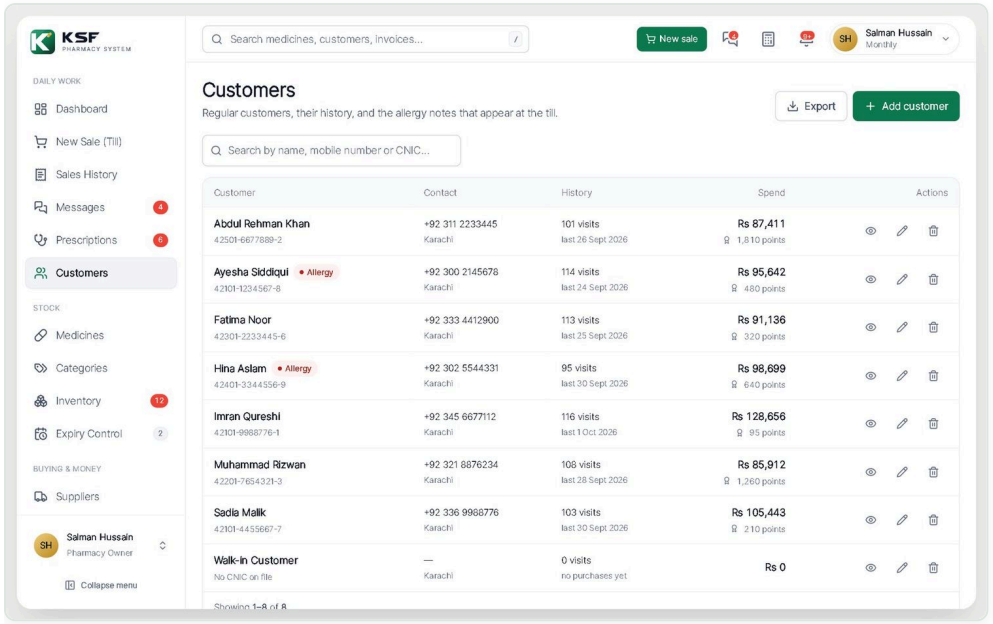
NEW · 2.0



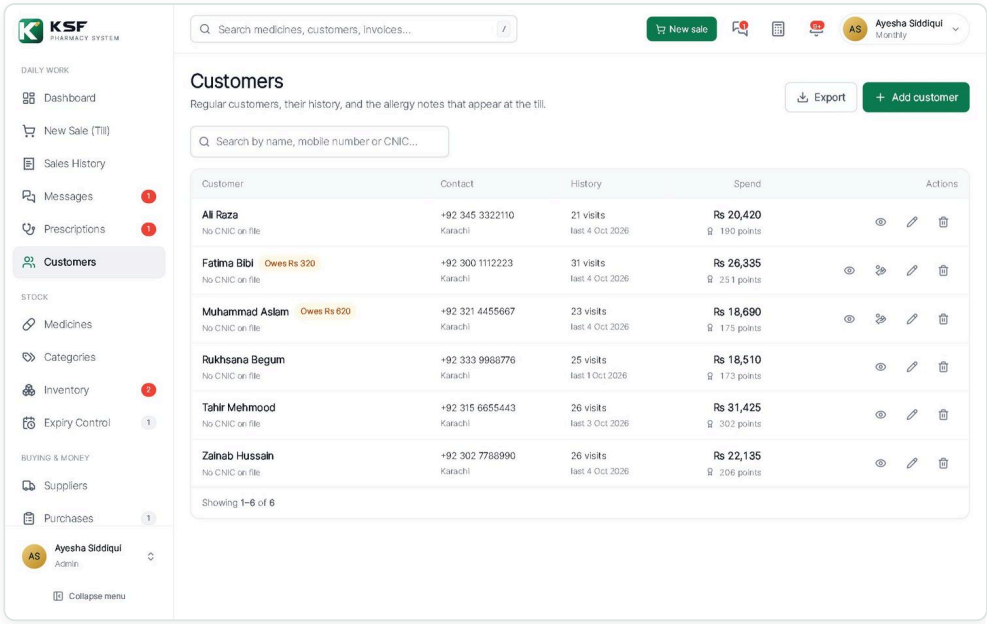
Same screen and filters.

Customers

PREVIOUS · 1.0.6



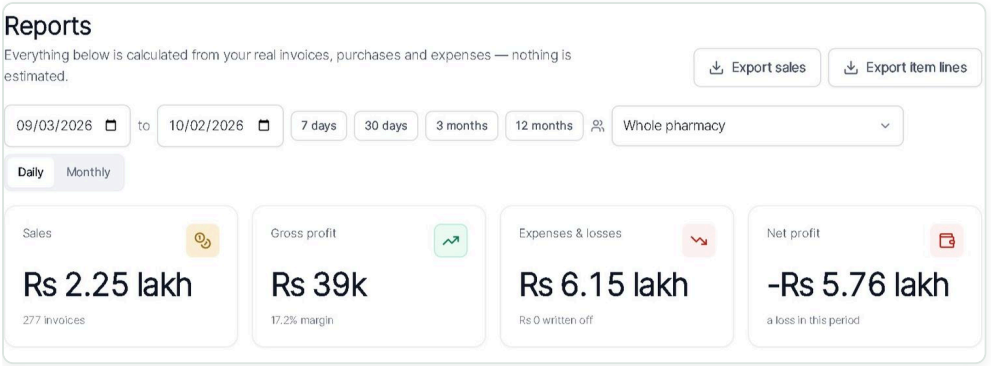
NEW · 2.0



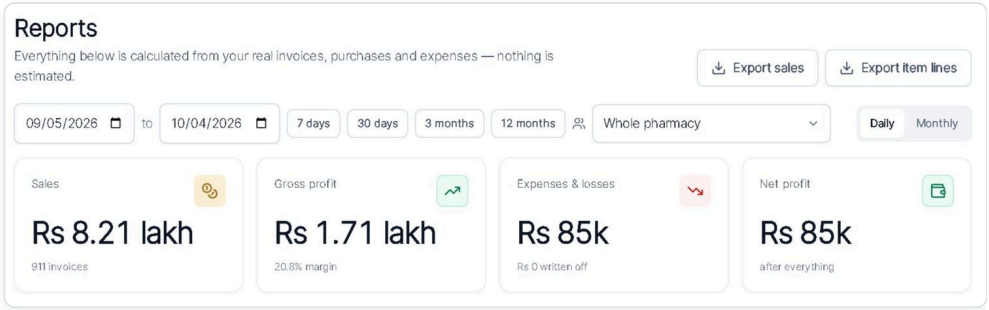
Same screen.

Reports

PREVIOUS · 1.0.6



NEW · 2.0



Same report and the same four headline figures; the numbers differ because the data differs. Below them, a 30-day chart now labels every third date so the dates no longer run into each other.

12 Small things found and fixed while testing

Taking the screenshots for this report is also a test. These came up and were fixed before hand-over.

What we saw	What it is now
Re-confirming the password for a sensitive action could sign the person out.	KSF asks for the password, then carries on with what the person was doing.
In the day close, a reversed udhaar repayment could be taken off the expected cash twice.	Counted once.
The "more actions" menu in the last rows of a table was cut off by the table edge.	The menu opens on top of the page, fully visible.
On a phone, a card with buttons in its header squeezed its title to one word per line.	The buttons move below the title on narrow screens.
On a 30-day chart the dates ran into each other.	Every third date is shown (every sixth on a phone); each bar still shows its value.
Two-column forms on My account were squeezed into narrow columns.	They use the full width of the card.

13 How this release was checked

Every check below can be run again with the command next to it.

Check	Command	Result
TypeScript, website and server	npm run typecheck	No errors
Automated tests (logic and the whole API against a real MongoDB)	npm test	49 of 49 pass
End-to-end: sign-up to day close, the platform console, support mode, import and undo, suspension	separate script, over HTTP	155 of 155 pass
Production build, with a final check of the bundle	npm run build	Pass — no secret in the website code
Each screen in this report	opened in Chrome	Pass at phone, tablet, laptop and wide-monitor sizes

The API tests run in a throwaway database that is deleted afterwards, so they never touch real data. They cover sign-up with an e-mail code, the till, refunds, udhaar, day close, reports, staff and roles, messages, themes, support requests, the platform console, support mode, data import and undo, keeping each pharmacy's records separate, suspension, and the Super Admin changing his own e-mail and password.

Known limits

- **Business days are counted in UTC.** A sale between midnight and 5 am Pakistan time counts towards the previous day in reports and the day close.
- **Online payment gateways** (JazzCash, Easypaisa, a card gateway) need your merchant account details and a live test before they are switched on. Bank transfer works now and is confirmed by the Super Admin.
- **E-mail** needs SMTP settings for a live system. Until then, sign-up codes appear in the server window.

14 The project folder

A clear split between the website, the server and everything around them. Files that were only there for the old demo, the old databases or earlier hand-overs have been removed.

Folder or file	What is in it
frontend/	The website and the application (React, TypeScript). One file per screen; platform/ is the Super Admin console
backend/	The server (Express, TypeScript): sign-in, the API by area, payments, e-mail, data import, database commands
tests/	Automated tests for the logic and the whole API
deploy/	Running on a server with Docker: the app, Nginx with HTTPS, optional MongoDB
windows/	Double-click launchers: start, check, back up, restore, build
docs/	Architecture, every setting, backups, payments, brand guide, and this report
.env	Your settings: the MongoDB link, the secret keys and the Super Admin's first sign-in
README.md	Overview, quick start, commands and the security model
DATABASE_SETUP.md	MongoDB step by step: Atlas, your own computer, or Docker; first start; backups; troubleshooting
CHANGELOG.md	What changed in each version

15 Before going live

A short list for the day KSF is opened to real pharmacies.

1. MONGODB_URI points at the live cluster; Atlas backups are switched on.
2. New secret keys for the live system: npm run secrets -- --write.
3. NODE_ENV=production, COOKIE_SECURE=true, HTTPS in front, and PUBLIC_APP_URL set to the real web address.
4. SMTP e-mail settings filled in, and one test sign-up completed with a real inbox.
5. The Super Admin's e-mail and password changed, and two-step sign-in switched on.
6. npm run db:backup scheduled every day, and one restore tried once.

deploy/README.md describes a full server installation with Docker, and docs/HANDOVER_REVIEW.md records exactly what was verified for this release.